Funding Proposal

FP128: Arbaro Fund – Sustainable Forestry Fund

Multiple Countries | MUFG Bank, Ltd | Decision B.25/15

15 April 2020

GREEN CLIMATE FUND

Funding Proposal

Project/Programme title:	*Arbaro Fund – Sustainable Forestry Fund.*
Country(ies):	*Sub-Saharan African and Latin American countries as follows: Ethiopia, Ghana, Sierra Leone, Uganda* *Ecuador, Paraguay, Peru*
Accredited Entity:	*MUFG Bank.*
Date of first submission:	*(2019/06/26)*
Date of current submission	*(2020/02/17)*
Version number	*[V.019 FINAL]*

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-MUFG]-[Country/Region]-[YYYY/MM/DD]"

A

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Programme	**A.2. Public or private sector**	Private
A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Mobilizing Funds at Scale (MFS)		
A.4. Result area(s)	*Check the applicable GCF result area(s) that the overall proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of GCF budget devoted to it. The total of the percentages when summed should be 100%.* Mitigation: Reduced emissions from: ☐ Energy access and power generation: ☐ Low-emission transport: ☐ Buildings, cities, industries and appliances: ☒ Forestry and land use: Adaptation: Increased resilience of: ☐ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☐ Infrastructure and built environment: ☐ Ecosystem and ecosystem services:	GCF contribution: 0% 0% 0% 100% 0% 0% 0% 0%	
A.5. Expected mitigation impact	***20 million tCO_2***	**A.6. Expected adaptation impact**	
A.7. Total financing (GCF + co-finance)	Up to 200 million USD	**A.9. Project size**	Medium (Upto USD 250 million)
A.8. Total GCF funding requested	25 million USD *For multi-country proposals, please fill out annex 17.*		
A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☐ Grant - ☐ Loan ☐ Guarantee	☒ Equity 25 million USD ☐ Results-based payment	
A.11. Implementation period	13 + 1 years	**A.12. Total lifespan**	15 + 1 years
A.13. Expected date of AE internal approval	*This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.* *Click or tap to enter a date.*	**A.14. ESS category**	*Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.* A
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☐ No ☒	**A.18. Is this FP included in the country programme?**	Yes ☐ No ☒

A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☐ No ☒
A.20. Executing Entity information	The Fund Manager FiM Asset Management S.à r.l. – AIFM, registered in Luxembourg ("AIFM" or the "Manager") and Arbaro Fund SLP, registered in Luxembourg as a special limited partnership ("Arbaro Fund" or "Arbaro") and Arbaro I GP S.à r.l ("General Partner")

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

Arbaro Fund is ranked top 30 for the request for proposal (RFP) – mobilizing funds at scale (MFS) out of 350 funding proposals. Arbaro Fund will provide effective climate change mitigation outcome through investing in sustainable plantation forestry projects in emerging forestry markets of Latin America and Sub Saharan Africa, while also bringing adaptation co-benefits. This investment approach provides developing countries and their rural communities with a solution to increase carbon sinks by producing wood in a sustainable manner and conserving natural forests, whilst contributing to reduction of illegal logging. Besides direct mitigation achieved in forest plantations, Arbaro operations will lead to additional GHG emission reduction benefits: the protection of natural forests in project areas, as well as the provision of alternative wood resources, will support efforts to combat deforestation and degradation of natural forests in project regions, and part of the carbon contained in harvested trees will be stored in wood products. These additional benefits are, however, not claimed by the Arbaro Fund.

The Arbaro Fund will identify and execute projects, focused on sustainable timber production based on the establishment of Greenfield projects and on the expansion, upscaling and technical improvement of Brownfield projects. Arbaro's average investment size will be between USD 10 to 25 million per project, with a final portfolio of 8 to 12 underlying investments / portfolio companies. In Latin America, initially Arbaro contemplates investments in Ecuador, Paraguay and Peru; in Sub-Saharan Africa in Ethiopia, Ghana, Sierra Leone and Uganda.

The key objectives of the Arbaro Fund are:

1) *Provide effective climate change mitigation*

When trees are planted on non-forested land, they capture and store additional carbon dioxide from the atmosphere. During the lifetime of Arbaro, through its activities in its projects Arbaro is estimated to sequester up to 20 million tonnes of CO_2 (net effect).

2) *Close the sustainable timber supply gap*

Most of the Arbaro's target countries face or will in the near future face a significant supply gap in sustainably produced industrial roundwood. The local industries utilising this raw material will face severe problems unless the prevailing development can be turned around. Arbaro's projects will seek to start closing this supply gap, and provide locally produced and sustainably produced timber, increasingly replacing imports and improving the country's trade balance.

3) *Support conservation efforts*

All plantation developments of Arbaro will be Forest Stewardship Council ("FSC") certified[1], which is still rare in the targeted regions (See page 6 – Climate Eligibility Criteria and Additionality). The approach includes an element of conservation of existing adjacent natural forest, wetland or other such vulnerable habitats. On average these FSC certified plantation forest project will set aside 20% of its total land area for conservation purposes. Hence it is expected that for every 1 000 planted hectares 200 hectares will be actively conserved.

By providing sustainably produced raw material to the local processing industries and forest product markets, Arbaro through its investments reduces the pressure to further increase utilisation of natural forest resources.

In addition, and following its investment strategy which requires provision of positive social impact in addition to economic and positive environmental impact, the Arbaro Fund contributes the following co-benefits

1) *Provide climate change adaptation co-benefits*

Improving resilience: By adaptive management practises (e.g. correct species and rotation choices) plantations can contribute to reduced vulnerability of near-by natural forests and other natural ecosystems by soil conservation, improved watersheds and by providing ecological corridors.

[1] For details please refer to B1 on Permanence and Annex 23

Providing alternative livelihoods: Smallholder farming is an important source of employment and income for rural populations. Providing alternative livelihoods through formal employment opportunities improves the resilience of the rural communities and decreases dependence on climate sensitive subsistence practices. Up to 200 000 rural people are expected to benefit from the investments.

2) *Empower local communities and enhance capacities*

Arbaro aims to promote out-grower schemes, where communities in the proximity of a forest plantations will be trained to apply modern silviculture to their own land, thus providing several benefits such as additional and diversified income, economic empowerment and improving rural capacities. Out-grower schemes are envisaged to be supported as a combination of technical assistance provided by Arbaro's investee projects and by grant funding programmes.

3) *Provide formal employment opportunities*

Arbaro, through its investments, is estimated to provide direct employment to ca 5 000 people. In addition, it is assumed that indirectly Arbaro's investee projects will employ another 15 000 people and provide improved livelihoods for up to 200 000 people.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate rationale and context (max. 1000 words, approximately 2 pages)

Climate rationale: Effective climate mitigation through development of new permanent carbon sinks

When trees are planted on non-forested land, they capture and store additional carbon dioxide (CO_2) from the atmosphere. In sustainable forest plantations[2] the forest is created over time and once the first planted trees reach their harvesting age, they are harvested but new trees are planted to replace them. This cycle is continued perpetually. With such forest management a permanent carbon stock is created[3], whilst providing raw material to the markets.

Arbaro Fund targets creating 75 000 hectares of new forest areas which will sequester up to 20 million tonnes of CO_2 during the lifetime of the programme. Arbaro Fund follows the Forest Stewardship Council ("FSC") for forest management, which requires, among other things, that new forests are planted in non-forest areas or areas which have been deforested long time ago[4] and therefore Arbaro Fund will never replace or convert existing natural forest or other vulnerable ecosystem with forest plantation.

Methodological approach

To estimate the carbon sequestration resulting from the implementation of the programme, Arbaro has developed a robust methodology that builds on internationally recognized approaches. First and foremost, Arbaro follows the general guidance of the IPCC[5] as described in the 2006 Guidelines for national GHG inventories and the 2003 Good Practice Guidance for LULUCF that provides sector-specific recommendations. The 2006 Guidelines are applied by more and more countries in their national inventories and reports to the UNFCCC and supplemented by the 2013 Good Practices Guidance as well as the 2019 Refinement, it will continue to become the standard approach to be used by all parties under the Paris Agreement. Following this guidance, the carbon stock changes in the project areas since the beginning of implementation of each project are monitored annually. Only those carbon pools that are changing considerably due to the implementation of the project are selected. In a forest plantation most of the carbon changes happen in the tree biomass, mainly in the stem but also in the branches, leaves and roots, however carbon is also stored in dead organic matter and the soil. To estimate the sequestration of carbon in a project, the timber volume in the plantations is measured first hand through forest inventories annually[6]. Stem volumes are thereafter converted into tCO_2 based on approved conservative conversion factors.

Arbaro only claims net carbon sequestered in forest plantations that is additional and permanent. To do this Arbaro uses approaches from the methodologies of the CDM, as well as the VCS and the Gold Standard, the most widely recognized standards for voluntary carbon certification of projects. The baseline scenario (= without Arbaro project scenario) is assessed and any potential sequestration in the baseline and sequestration achieved before the programme activities is deduced from the sequestration achieved through programme activities. Any emissions that may result from the implementation of the programme is accounted and deducted from the programme sequestration. All harvested wood is assumed as a loss and no carbon stored in harvested wood products is claimed. To account for losses due to harvest a maximum sequestration that can be claimed is the long-term average, which is set using the recognized methodology:

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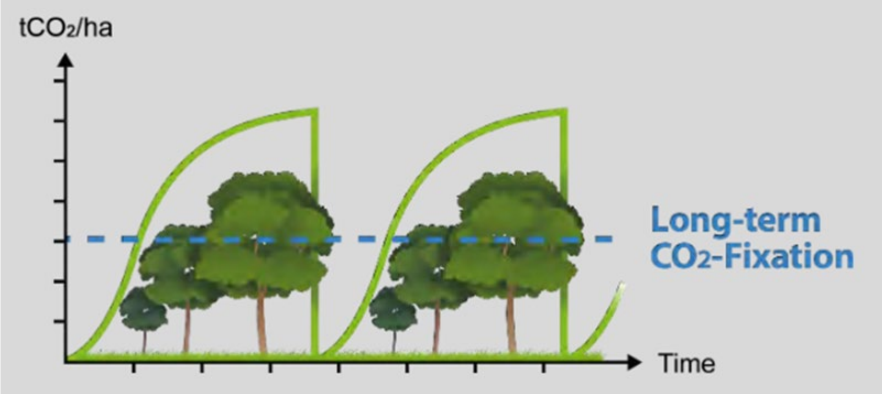
[2] Please refer to Annex 23 on sustainable forest plantations
[3] Please refer to Annex 22 on carbon methodologies
[4] Cut off of 1994
[5] Intergovernmental Panel on Climate Change - the United Nations body for assessing the science related to climate change
[6] Forest inventories, which provide the first step in the mitigation impact calculation are based on scientifically approved methodologies and protocols and procedures accepted and globally used in the forestry sector. These consist of the physical, on the ground measurement of trees in representative forest samples, which allows to calculate the estimated existing volume in the plantations and growth over time. These data must be kept in the portfolio company records and, in the case of FSC be available for FSC to study, and in the case of Arbaro, be available for Arbaro fund to access and monitor.
Arbaro fund monitors these annually, and these are used in the calculation of the carbon sequestration.
The carbon sequestration results (among other items) are reported to the GCF as part of the annual reporting cycle.



Source: Gold standard

Permanence

Permanence is a concern in reforestation projects. Arbaro follows the approach of the VCS and Gold Standard. These certification methods issue permanent carbon credits and thus need to ensure that GHG benefits are permanent. For this they give great importance to the long-term sustainability of the projects: projects which are economically viable, socially beneficial and environmentally sound can continue operations in the long term. Both standards consider FSC certification as a proof of sustainability. Following this approach, all of Arbaro' s projects must achieve FSC certification.

FSC was founded in 1993 through joint effort of environmental and social NGOs as well as producers and retailers of wood products. The FSC is a global multi-stakeholder membership scheme with a strong emphasis on NGO participation. FSC requires adherence to principles and criteria which need to be followed in a manner appropriate to the scale, intensity and risk of the project. FSC certification does not automatically last forever. The certified projects go through annual audits by external auditors and in addition, the certification has to be renewed every five years.

The FSC principles[7] are:

- Principle #1: Compliance with laws - The certified entity shall comply with all applicable laws, regulations and nationally-ratified international treaties, conventions and agreements.
- Principle #2: Workers' rights and employment conditions - The certified entity shall maintain or enhance the social and economic well-being of workers.
- Principle #3: Indigenous peoples' rights - The certified entity shall identify and uphold indigenous peoples' legal and customary rights of ownership, use and management of land, territories and resources affected by management activities.
- Principle #4: Community relations - The certified entity shall contribute to maintaining or enhancing the social and economic well-being of local communities.
- Principle #5: Benefits from the forest - The certified entity shall efficiently manage the range of multiple products and services of the Management Unit (=certified area) to maintain or enhance long term economic viability and the range of environmental and social benefits.
- Principle #6: Environmental values and impact- The certified entity shall maintain, conserve and/or restore ecosystem services and environmental values of the Management Unit, and shall avoid, repair or mitigate negative environmental impacts.
- Principle #7: Management planning - The certified entity shall have a management plan consistent with its policies and objectives and proportionate to scale, intensity and risks of its management activities. The management plan shall be implemented and kept up to date based on monitoring information in order to promote adaptive management. The associated planning and procedural documentation shall be sufficient to guide staff, inform affected stakeholders and interested stakeholders and to justify management decisions.
- Principle #8: Monitoring and assessment - The certified entity shall demonstrate that progress towards achieving the management objectives, the impacts of management activities and the condition of the

[7] Please refer to Annex 23, or www.fsc.org

Management Unit, are monitored and evaluated proportionate to the scale, intensity and risk of management activities, in order to implement adaptive management.

- Principle #9: High conservation values - The certified entity shall maintain and/or enhance the high conservation values in the Management Unit through applying the precautionary approach.
- Principle #10: Implementation of management activities- Management activities conducted by or for the certified entity for the Management Unit shall be selected and implemented consistent with the certified entity's economic, environmental and social policies and objectives, and in compliance with the Principles and Criteria collectively.

In addition, there are country specific adaptations of these principles and criteria. Today, FSC is the most recognized environmental and social certification scheme for forest management and known for its stringent requirements including annual external audits for certified projects.

Additional mitigation benefits through conservation and land use

Creating sustainable forest plantations also means that any existing natural forests or areas with otherwise high conservation value, such as wetlands, which are part of the overall project area, are conserved. On average this means that between 20% of the overall project area is conserved, with the responsibility of maintaining the integrity of these areas being placed on the project entity. These conservation areas provide further mitigation benefits, which however are not claimed as part of the programme. Together the plantation area and the conserved natural areas increase the resilience of the overall landscape against climate change impacts. Finally, in case of Arbaro Fund's projects, majority of the foreseen products produced from the wood will be mechanical forest products such as panels or sawn timber for e.g. furniture manufacturing or construction. In these products the carbon stock continues throughout the lifecycle of the product, i.e. significantly beyond harvesting of the tree. This additional sequestration is not claimed by the Arbaro Fund.

Arbaro Fund, by afforesting or reforesting oftentimes degraded land and protecting attached conservation areas of natural forest, will both establish new carbon sinks and contribute to reduced emissions from deforestation and land use change. This will make an important contribution to enable the target countries to achieve their climate targets, as forestry is one of the few sectors that allows sequestration of carbon dioxide and is reflected in many INDCs to the Paris Agreement and the national climate strategies. For detailed analysis of the target countries' climate policies and regional circumstances, please refer to Section D5 and Annex 23, Climate rationale, country context.

Climate eligibility criteria and additionality

Arbaro Fund is governed by its investment strategy, climate eligibility criteria and its ESMS. Three key building blocks of the investment strategy are:

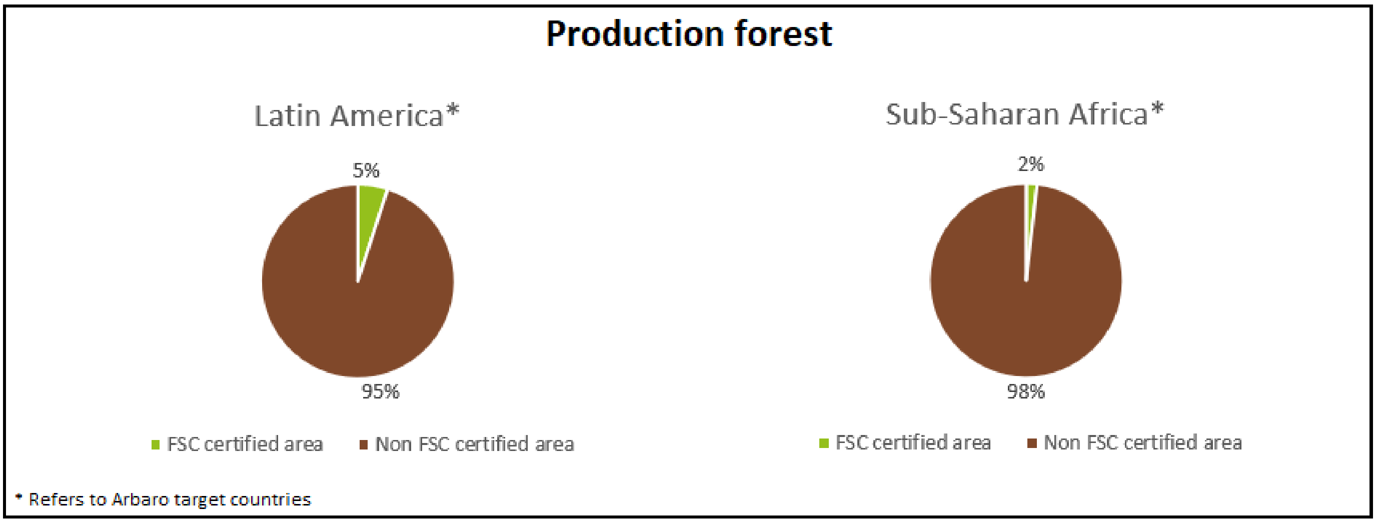
1) Climate mitigation benefits of the projects and their ability to provide effective carbon sequestration
2) Long term financial viability of the projects and their ability to provide returns to the investors; and
3) Social co-benefits of the projects and their ability to provide formal employment in their areas of operation.

As part of the climate eligibility, Arbaro assesses the baseline, barriers, investment case, funding mobilization potential as well as mitigation benefits (net sequestration impact) of the proposed project together with co-benefit creation potential. These together with the framework of conditions derived from the investment strategy and ESMS determine whether the proposed project will proceed in Arbaro's investment process[8].

In Arbaro's target countries, FSC certified forests are still rare. The forest area with FSC certification represents a share of 5% in Latin America and 2% in Africa. These numbers include all production forests, also natural forests which tend to be much larger scale than forest plantations. Forest production under FSC standards is not a common practice in Arbaro countries, but an innovative approach that needs substantial support to become the industry standard and contribute to the shift towards sustainable production in the sector.

[8] Climate eligibility criteria of the Arbaro Fund

Figure 1 FSC certified forests in Arbaro's target countries in Latin America and Sub-Saharan Africa



Responding to increased demand through sustainable production

Demand for wood-based products will grow to unprecedented levels over the coming decades[9]. By 2050, the world's population is projected to reach 9.7 billion. Rising numbers of people are only part of the story. At the same time, economic growth, urbanization and increasing prosperity are driving greater consumption[10]. Meeting these growing demands will increase the already huge pressure on the world's natural resources, forests and wood being no exception. WWF's Living Forests Model projects that wood harvesting will more than double by 2030, and almost quadruple by 2050: from 3.4 million m^3 in 2010, to 7.6 million m^3 in 2030 and 13.1 million m^3 by mid-century[11]. There are various measures that societies could, and should, employ to reduce the need to harvest trees – including more efficient saw mills and pulp mills, greater recycling, less wasteful consumption in rich countries and energy-efficient improved wood stoves in developing ones. But these will have only a limited impact. The bottom line is that we will be using more wood than ever before and there are limits to how much wood can be harvested from natural forests. Currently, planted forests make up only around 7% of forest cover worldwide, but supply a third of the total global production of industrial roundwood. Sustainable forest plantations can therefore provide an effective tool to produce the wood in smaller land areas efficiently.

In addition to having excellent bio-physical and general climatic conditions for growing forests, Arbaro's target countries are characterized by the macroeconomic drivers that fuel consumption:

2019-2024	Ethiopia	Ghana	Sierra Leone	Uganda	Ecuador	Paraguay	Peru
Population growth	2.49%	2.09%	2.03%	2.86%	1.30%	1.19%	1.04%
Economic growth	6.89%	4.90%	4.69%	7.25%	1.97%	4.05%	3.85%

Compound annual growth rates projected by the UN (population) and the IMF (real GDP)

Barriers and gaps

As an asset class, sustainable forest plantation investments in emerging markets and developing countries is unknown to most investors and particularly private investors have so far not been very active in the sector. DFIs/IFIs are typically among the frontrunners for new asset classes in less developed markets and can, as anchor investors, demonstrate confidence in new asset classes. However, in the case of plantation investments even the DFIs

[9] WWF 2012
[10] Kharas 2017
[11] WWF 2012

involvement has been limited so far. Patient capital and risk-taking capital – both public and private – is necessary to create evidence and catalyze bigger investments into the sector in developing countries. Hence, a successful launch of the Arbaro Fund has the potential to prove the environmental, social and economic concept of commercial sustainable forestry plantations and stimulate further – urgently needed – investment into the sector. It can contribute to bringing climate change mitigation through forestry from a broad concept on paper to scalable and replicable on-the-ground climate action.

In climate change mitigation and adaptation, sustainable plantation forestry programmes have not been widely used with Renewable energy and REDD and REDD + programmes having taken a more significant role receiving more attention from public and private investors. The underinvestment in Arbaro's target countries in this sector can be clearly seen in the reforestation rates these countries suffer but also in the proportion of sustainably produced plantation grown raw materials being available (see Table 3). Sustainable plantations contribute to the overall shift towards truly sustainable production of wood and wood-based materials with significantly smaller land area requirements than natural forest logging whilst providing significant additional sequestration benefits and contributing to keeping the existing forest carbon stocks of natural forests untouched.

Adaptation co- benefits

Providing positive social impact is in the core of Arbaro Fund's investment strategy, and one of the three key target deliverables that Arbaro has. It is also emphasized in the investment guidelines and in the ESMS of Arbaro. All of these are part of the binding legal documentation of Arbaro. Arbaro fund's projects always have to achieve FSC certification, which also requires the certified organization to maintain or enhance the social and economic well-being of local communities as well as employees.

In addition to provision of formal employment, the type of other co-benefits varies from project to project and depends on the local conditions and the needs of the local communities and other stakeholders. The overall social co-benefits of the Arbaro Fund are discussed in more detail in section D3, however below is a short summary.

Many economies in Latin America and Sub-Saharan Africa still strongly depend on agriculture. The national development plans and climate strategies of the target countries aim to reduce the vulnerability of the most exposed communities. Currently, about 40% of the world's working age population lives in rural areas. The rural economy, which includes nearly 80% of the world's poor workers, is characterized by serious decent work deficits. Labour market, organizational and representative institutions tend to be weak. Underemployment is widespread, incomes are low and access to social protection is limited[12].The distinctive characteristics of rural areas make them uniquely vulnerable to the impacts of climate change because of greater dependence on agriculture and natural resources makes them highly sensitive to climate variability, extreme climate events, and climate change (IPCC). Furthermore, low income base and lower than average education level increases this sensitivity as there is limited capacity to wait and limited alternatives.

By providing alternative livelihoods in rural areas through formal employment opportunities, sustainable forest plantations are key to increasing the resilience of both ecosystems and vulnerable populations facing the threats of climate change. Arbaro's projects will grow longer rotation logs for e.g. sawlogs and veneers logs, for panels and constructions material, ILO statistics show that longer-rotation solid wood plantation industry makes significant job contributions in contrast to short-rotation pulp and paper productions. Formal employment opportunities also decrease pressure to migrate to urban areas. Finally, formal employment creates economic activity, which in turn creates further alternative livelihood options and decreases dependence on subsistence practices such as subsistence agriculture or charcoal production and the vulnerability of the communities.

Profit sharing

Arbaro prefers land lease structures to land ownership, where possible, in order to allocate more funding to afforestation rather than purchasing land. When such land lease structures are applied, benefit sharing mechanisms are put in place to assure long term alignment between the stakeholders (e.g. landowners, communities). These structures vary from project to project but can include, in addition to annual land lease being paid, a royalty payment on volumes harvested or other such profit share mechanisms.

[12] ILO 2016; Working in Rural Areas in 21st century

Community programmes

In addition to provision of formal employment, the forestry projects that Arbaro supports, provide multiple benefits to the local communities. The form of these benefits varies between the projects and depends on the needs of the communities. These projects can include helping the communities with water sources (bore holes), building of school rooms, providing scholarships to local children, providing technical assistance to agriculture projects to improve food production, outgrower schemes to grow community forests, or building or repairing of roads or bridges. Below we provide a short case example of Arbaro's first investment, to highlight the social impact that the company has undertaken.

Case example

Arbaro's first investment project, for which Arbaro provides expansion capital, works closely with the local communities in its areas of operation in Sierra Leone and Ghana. The company provides direct employment to 1 600 people of which 17% are female. All employees have access to medical cover, maternity leave, compassionate leave and sick pay. The company pays salaries that are 30 % higher than the national minimum wage, and employees receive regular training and skills development. Employees are hired locally unless individuals with the required skillset are not available. The additional jobs in the region reduce the strain of communities on subsistence agriculture and land pressure. In many cases where subsistence agriculture fails (from e.g. climate related reasons) there are no other income sources, making the employment provided by this project crucial.

All employees and stakeholders can easily report possible compliance violations through an accessible grievance mechanism and confidential whistleblowing policy.

Training and knowledge sharing programmes reach beyond the immediate employees and cover also the communities in the proximity of the company's operations and other stakeholders. In 2018 the company's health awareness program reached more than 3000 people in the communities. The company also regularly arranges community safety campaigns in e.g. road safety and fire prevention.

The company contributes to improving the local infrastructure, and has to date upgraded, built or repaired approximately 2 000 kilometers of roads or rural trails.

The company supports community projects, which are chosen based on the communities' needs. The projects completed so far include construction of six school blocks in the communities, education support to children, through scholarships and provision of school materials - 2,000 children have benefited from such programs, reparation and construction of wells to facilitate access to clean water and agriculture projects, with ten communities per year, where the company provides inputs and technical support, as well as capacity building in agricultural activities and saving profits for future planting seasons.

Figure 2 Pictures from community meeting and work place training exercise

Figure 3 School building built by the company as part of their community programme

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

Arbaro's target countries face or will face a significant supply gap in sustainably produced industrial roundwood in the near future. The wood products produced from this raw material include panels, sawn wood, as well as various applications of these such as furniture, pallets and various construction materials such as doors, window frames and roof trusses. This means, that these products either need to be sourced from unsustainable (and often illegal) sources, imported or replaced by less sustainable products (e.g. using aluminium roof trusses instead of wooden ones). In these markets, which from biophysical perspective, could produce these products locally, such developments are particularly worrying.

This gap is expected to further widen in the coming years as these economies are rapidly growing. This coupled with population growth, the two being the key drivers for wood product demand, will lead to expanding gap and increased pressure towards the already vulnerable remaining natural forests.

Table 1 – Country data

	Ethiopia	Ghana	Sierra Leone	Uganda	Ecuador	Paraguay	Peru
Sustainable supply gap[13]	4-5 million m³	2-3 million m³	1 million m³	3 million m³	1.0 million m³	1 million m³	2-3 million m³
Population growth (2019-2024)	2.49%	2.09%	2.03%	2.86%	1.30%	1.19%	1.04%
Economic growth (2019-2024)	6.89%	4.90%	4.69%	7.25%	1.97%	4.05%	3.85%

Compound annual growth rates projected by the UN (population) and the IMF (real GDP); sustainable supply gap estimated by Arbaro Advisors

Arbaro's projects will make a significant contribution to CO_2 sequestration: When trees are planted on non-forested land, they capture and store additional carbon dioxide from the atmosphere. Since the focus of Arbaro is on sustainable timber production from planted forests, Arbaro conservatively only claims additional and permanent carbon sequestration achieved through afforestation / reforestation and carbon stock enhancement in planted forests through improved forest management[14].

[13] Annual industrial roundwood demand being met from unsustainable sources
[14] See Annex 22 Carbon methodologies

At the same time, Arbaro's projects will see to start closing the supply gap, and provide locally sourced and sustainably produced timber, contributing to moving the local forestry sector towards sustainable sourcing. The wood produced in Arbaro's projects will replace unsustainable and informal sources of natural forest wood or imports and improving the country's trade balance. Arbaro's projects will also serve as an example for further similar developments in these countries.

The local industries utilising the wood raw material will face severe problems unless the prevailing development can be turned around. At the same time, unsustainable harvesting of remaining natural forests deteriorates landscapes and reduces carbon sinks thus accelerating climate change. Also, unsustainable harvesting of natural forests tends to be done on an informal basis without adhering to legality requirements, paying taxes, duties or royalties and legislation and best practises on labour, health and safety. This contributes to reduced economic resilience of the rural regions and increased vulnerability of its people.

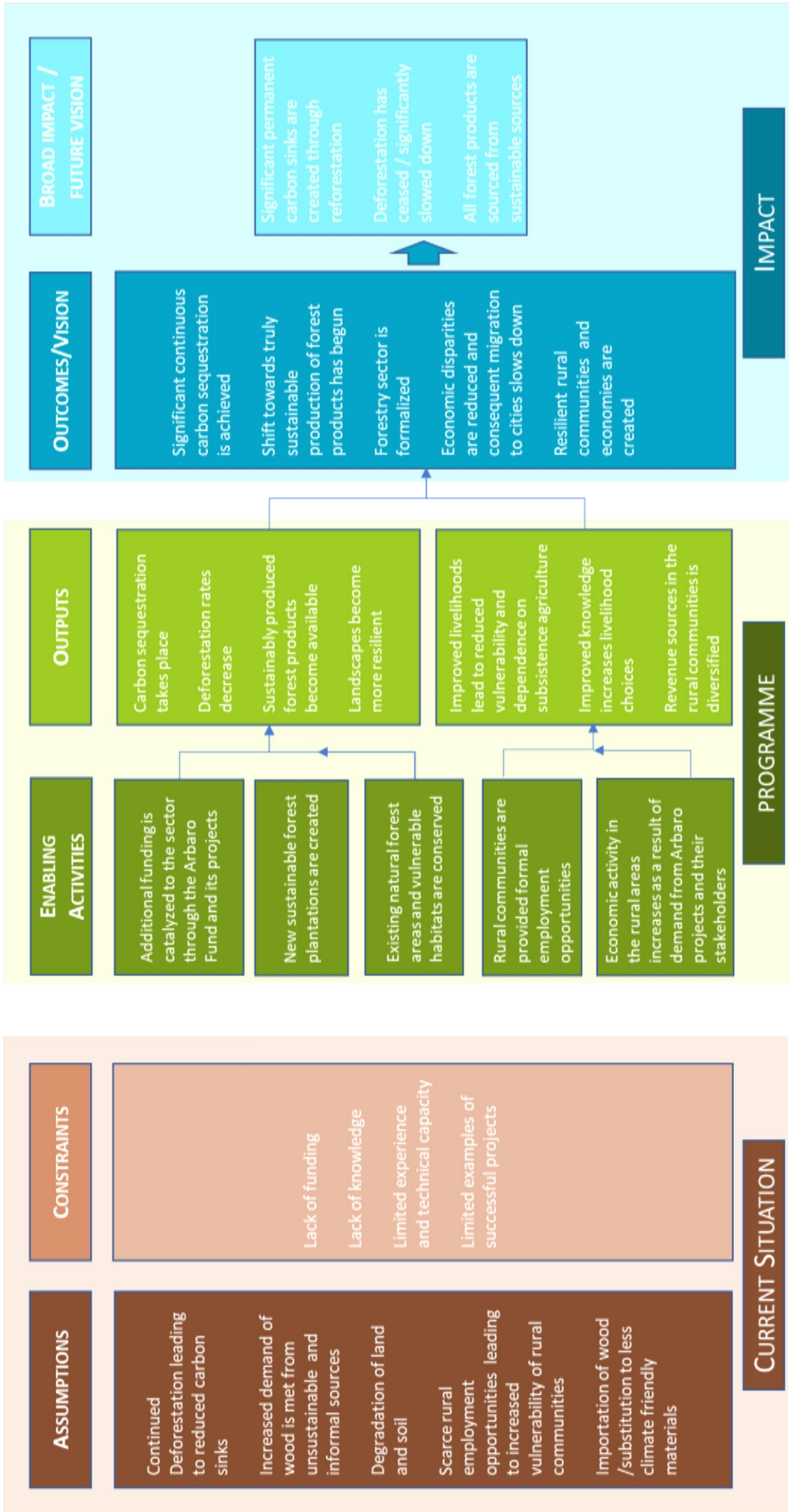
Arbaro's projects will also in most cases include value adding processing, leaving higher value of the production in the region / country and providing locally made products into the markets (rather than e.g. selling squared logs to exports to be then produced as final products). As an example, Arbaro's first project is currently producing veneer from its wood, soon entering the next step in value addition and starting to produce plywood. Plywood is a versatile product capturing a price per cubic meter of manifold to veneer. There is significant demand in the countries and nearby region for plywood, and with no sustainable supply currently available, this product is currently mainly imported or produced from unsustainable natural forest resource.

Arbaro's activities can provide the mitigation impact (sequestration) and the co-benefits (supply of sustainably produced timber, provision of formal rural employment, increased economic activity) when done in scale. Arbaro reached its first closing after being shortlisted in the RFP (Mobilizing Funds at Scale), with some investors taking the view that Arbaro will grow to scale. But the 60 million USD of the first closing is too small, and it is essential that the Arbaro Fund grows to a beyond 120 million level. That will provide long term sustainability of the Arbaro Fund management team, adequate diversification of projects and through these the example effect to these markets. GCF plays a crucial role in this scaling up. It is expected that with GCF investment of 25 million, the Arbaro Fund size will reach ca 140 million USD dollars, and potentially attract further investors to reach its full size of 200 million.

With this and the leveraging effect that Arbaro's equity investment will have on the project level, the overall multiplying effect of GCF investment would reach to 15x.

GCF funding is sought with matching terms with the other Arbaro Fund investors and with same ranking.

Arbaro's Theory of change diagram


ASSUMPTIONS
Continued Deforestation leading to reduced carbon sinks
Increased demand of wood is met from unsustainable and informal sources
Degradation of land and soil
Scarce rural employment opportunities leading to increased vulnerability of rural communities
Importation of wood /substitution to less climate friendly materials
CONSTRAINTS
Lack of funding
Lack of knowledge
Limited experience and technical capacity
Limited examples of successful projects
CURRENT SITUATION
ENABLING ACTIVITIES
Additional funding is catalyzed to the sector through the Arbaro Fund and its projects
New sustainable forest plantations are created
Existing natural forest areas and vulnerable habitats are conserved
Rural communities are provided formal employment opportunities
Economic activity in the rural areas increases as a result of demand from Arbaro projects and their stakeholders
OUTPUTS
Carbon sequestration takes place
Deforestation rates decrease
Sustainably produced forest products become available
Landscapes become more resilient
Improved livelihoods lead to reduced vulnerability and dependence on subsistence agriculture
Improved knowledge increases livelihood choices
Revenue sources in the rural communities is diversified
PROGRAMME
OUTCOMES/VISION
Significant continuous carbon sequestration is achieved
Shift towards truly sustainable production of forest products has begun
Forestry sector is formalized
Economic disparities are reduced and consequent migration to cities slows down
Resilient rural communities and economies are created
BROAD IMPACT / FUTURE VISION
Significant permanent carbon sinks are created through reforestation
Deforestation has ceased / significantly slowed down
All forest products are sourced from sustainable sources
IMPACT

B.3. Project/programme description (max. 2000 words, approximately 4 pages)

Introduction

Arbaro will make investments in sustainable plantation forestry projects / companies which will expand, upscale and technically improve brownfield projects and / or establish new plantations in selected countries in Latin America and Sub-Saharan Africa, in which the Arbaro team has a demonstrated track record and long term on the ground experience, as well as solid networks. The team's network has allowed Arbaro to build a strong investment pipeline. Total number of investee projects will be 8-12. Arbaro investments will be equity investments, or investment in equity-like instruments including but not limited to shareholder loans or mezzanine facilities. Final country allocation, exact location of the projects as well as number and scale of the project is yet unknown as the projects will be implemented over the investment period of the Arbaro Fund, which ends in July 2022. It is however expected that the Fund will allocate a minimum of 5% of its total volume to each target country (please refer to Annex 17).

Arbaro Fund activities:

1) *Pipeline development*

   The Arbaro Fund investment advisors will identify potential pipeline projects and screen those based on Arbaro's investment strategy, climate eligibility criteria and ESMS.
   Projects that based on initial assessment fulfill the criteria move forward in the investment process to full formal due diligence.

2) *Assessment of project opportunities following Arbaro's investment strategy and ESMS*

   The Investment Committee of the Arbaro Fund is involved in the various stages of the investment process. Investment Committee approval is required before the proposed project can move to the execution stage.
   As part of the execution of an investment, the investment documentation is negotiated and entered into, after which the investment can be made.

3) *Investments in project level operations*

   Arbaro invests in Portfolio companies and as part of the investment it is determined for what purposes the investment proceeds may be used. These include i.a. land rental or purchase, development of new plantation areas, management of forest areas, establishment of conservation areas, nursery operations, harvesting, transport and processing as well as other such portfolio company operations as approved by Arbaro.

4) *Arbaro Fund investment monitoring*

   At the Arbaro Fund level, after the investment execution the main function of Arbaro is to monitor the portfolio company operations through reporting from the companies as well as active dialogue, participation in the boards of the companies as well as doing regular on-site visits to the production sites. This is to ensure that the investment conditions, including the ESMS is adhered to, and the impacts and returns are delivered.
   For this, the performance of the portfolio companies is reviewed to assess performance against the targets, including financial, climate, environmental as well as social return and impact targets.

5) *Reporting throughout the Arbaro Fund lifetime*

   The investors of the Arbaro Fund will receive quarterly updates on the performance of the portfolio companies as well as an annual E&S and Impact report.

6) *Exit from investments*

   The Arbaro Fund has a limited lifetime of 15 years. Towards the end of this period the Arbaro Fund needs to start assessment of potential exit channels for each of the projects. This is done following the exit strategy of the Arbaro Fund (see section B.6.).
   After Investment Committee approval the exit process continues to negotiation and execution of all exit documentation.
   After execution, the exit proceed are received by the Arbaro Fund and subsequently distributed to the investors.

Investment Strategy of the Arbaro Fund

Investee projects may be new greenfield developments where the institutional structures need to be set up or existing enterprises in need of expansion capital. In each case however, there needs to be an element of increasing the forest cover and in the case of existing enterprises improving of forest management practices to provide the sequestration benefits.

In addition, Arbaro will invest a minor share of its capital in selected value enhancing downstream activities such as wood processing. These investments will not be made on a stand-alone basis, but always integrated with a plantation investment.

Arbaro will always cooperate with local partners. These may be for example:

- local landowners;
- local existing forestry businesses seeking to expand their operations or needing a raw material source but not having financial resources or knowhow to grow it;
- other off-takers needing the material (e.g. further value adding processors);
- local investors.

Therefore, Arbaro's projects are developed taking into account the local needs and the local markets.

Arbaro's target regions and the initial target countries have ideal biological growing conditions and a growing domestic demand for timber products. The strict application of the "Site-Species-Market" approach for all investments will ensure that Arbaro's projects have access to various markets and different types of off-takers. This approach involves careful site and species selection, the use of up-to-date silvicultural practices and thorough considerations of market access and marketing strategy. The availability of water will be particularly relevant for site - species matching in regard to annual rainfall and distribution, as well as potential changes resulting from climate change. This consideration, together with measures such as erosion control, proper land preparation techniques, and protection of riparian vegetation will minimize any potential adverse impacts on water resources. The consideration of these water matters is specifically required by several criteria of the FSC standard[15].

Arbaro prefers land lease models instead of land acquisition, which allows for optimal capital deployment to achieve a balanced investment portfolio with an optimized risk-return profile. A mix of mix of different rotations, as well as different species and age classes allows for more evenly spread out cash flows, interim capital redemptions and early as well as later exits.

All of Arbaro's investments will be FSC certified. Investments in social and environmental aspects are considered to be an integral part of each project to de-risk the projects to potential E&S issues and increase the quality and quantity of the production as well as its long-term sustainability.

Arbaro will actively manage the projects with its deep expertise in plantation management, with an aim to create strong companies and to realize the full potential of the identified plantations. Arbaro will inject management expertise in terms of planning, coordination and monitoring. Strong corporate governance systems will be implemented, as well as state-of-the-art forestry management systems.

[15] Principle 6: Environmental Values and Impacts. Criterion 6.7 The Organization shall protect or restore natural water courses, water bodies, riparian zones and their connectivity. The Organization shall avoid negative impacts on water quality and quantity and mitigate and remedy those that occur.

Principle 9: High Conservation Values. Criterion 9.1 The Organization, through engagement with affected stakeholders, interested stakeholders and other means and sources, shall assess and record the presence and status of the following High Conservation Values in the Management Unit, proportionate to the scale, intensity and risk of impacts of management activities, and likelihood of the occurrence of the High Conservation Values: HCV 4 - Critical ecosystem services. Basic ecosystem services in critical situations, including protection of water catchments and control of erosion of vulnerable soils and slopes.

Principle 10: Implementation of Management Activities. Criterion 10.2 The Organization shall use species for regeneration that are ecologically well adapted to the site and to the management objective. The Organization shall use native species and local genotypes for regeneration, unless there is clear and convincing justification for using others. Criterion 10.5 The Organization shall use silvicultural practices that are ecologically appropriate for the vegetation, species, sites and management objectives. Criterion 10.10 The Organization shall manage infrastructural development, transport activities and silviculture so that water resources and soils are protected, and disturbance of and damage to rare and threatened species, habitats, ecosystems and landscape values are prevented, mitigated and/or repaired.

Financial and return aspects

Arbaro's portfolio companies will derive their financial returns mainly from the sale of timber. Income for Arbaro is either derived from timber sales (dividends from portfolio companies) or through the sale of ownership stakes in the portfolio companies. By selecting the right investments, actively managing the projects, and ensuring an attractive exit, Arbaro targets an IRR of 12% p.a. for its investors. The implied gross return at project level is ca 15 %.

The three key return drivers in plantation forestry are the prices of end products, biological growth rates, and operational costs (see figure below). Arbaro will mainly target holding project land via lease agreements and therefore land prices also matter, but in most cases the impact on returns is smaller. Please note that each project is different and specific return drivers can vary widely. Before investment, Arbaro will always carry out a rigorous due diligence, financial modelling and benchmarking exercise to determine a wider array of value drivers as well as suitability of investment.

Initial target countries and markets

Arbaro focuses on emerging forestry markets in two main regional clusters within which the team will focus on selected countries. In Latin America, Arbaro will initially concentrate on Ecuador, Paraguay and Peru. In Sub-Saharan Africa, Arbaro will initially focus on Ethiopia, Ghana, Sierra Leone and Uganda.

Apart from strong macroeconomic drivers for increased wood products demand and good bio-physical growing conditions for forests, all of these countries are also characterized by deforestation which is partly driven by logging of natural forests:

Table 2 Deforestation drivers

Region	Country	Main deforestation driver	Logging a deforestation driver?	Share of primary deforestation attributable to logging
Sub-Saharan Africa	Ethiopia	Shifting agriculture	No*	<1%
	Ghana	Shifting agriculture	Yes	<10%
	Sierra Leone	Shifting agriculture	Yes	<10%
	Uganda	Shifting agriculture	Yes	<10%
Latin America	Ecuador	Shifting agriculture	Yes	<10%
	Paraguay	Large-scale agriculture	Yes	<10%
	Peru	Shifting agriculture	Yes	<10%

* Native forests of Ethiopia too degraded to allow for logging; however, wood fuel extraction is one of the most important drivers of forest degradation

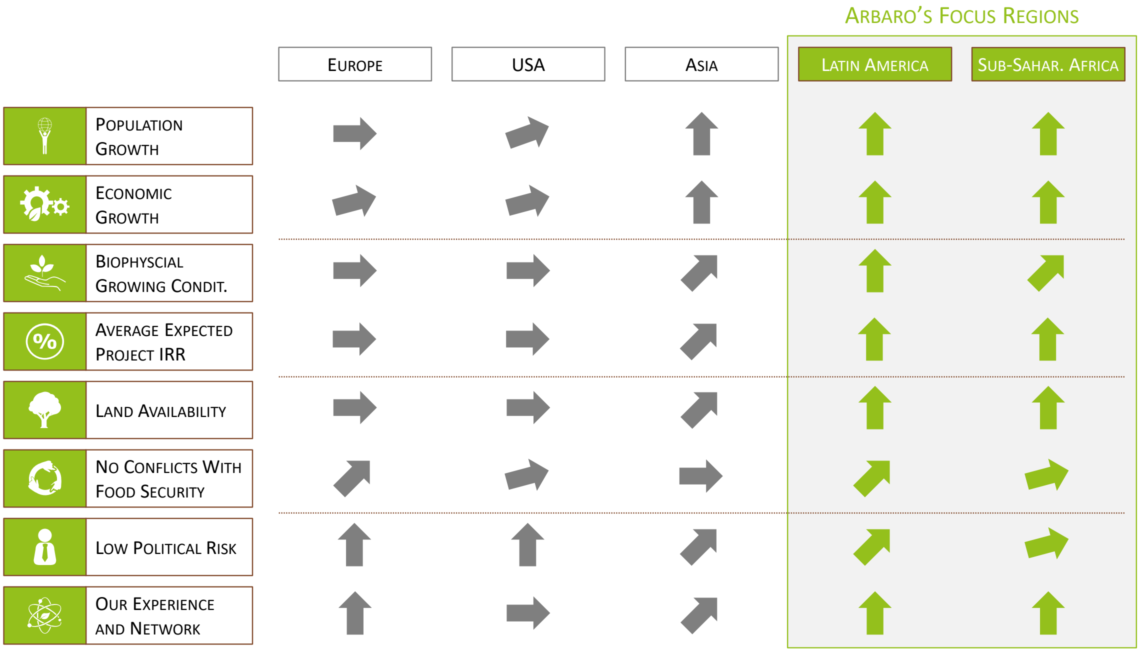
Please note that deforestation statistics by driver are not readily available. In all of the target countries the bulk of primary deforestation is due to agriculture. Logging and wood fuel extraction have been identified as primary deforestation drivers, but at a lower scale. However, more importantly, unsustainable wood removals are typically a main cause of forest degradation, i.e. among the key secondary deforestation drivers, which over time leads to easier clearing of the land area for agriculture.

In both regional clusters, the identified pipeline projects mainly produce timber for the domestic and regional markets.

Competitive advantages of target regions

Latin America is a more mature forestry region than Sub-Saharan Africa and has a higher share of timber exports. Land prices and labor costs are higher in Latin America than in Africa. However, the productivity of the land combined with a higher availability of skilled labor, up-to-date technology and professional service providers compensates for the higher costs. The major advantage of Sub-Saharan Africa is the in principle high availability of land, competitive land prices and very favorable domestic demand dynamics due to rapidly growing local populations and economies. In consequence, overall timber production costs differ more on project level than between the two target regions.

Figure 4 – Competitiveness of regions



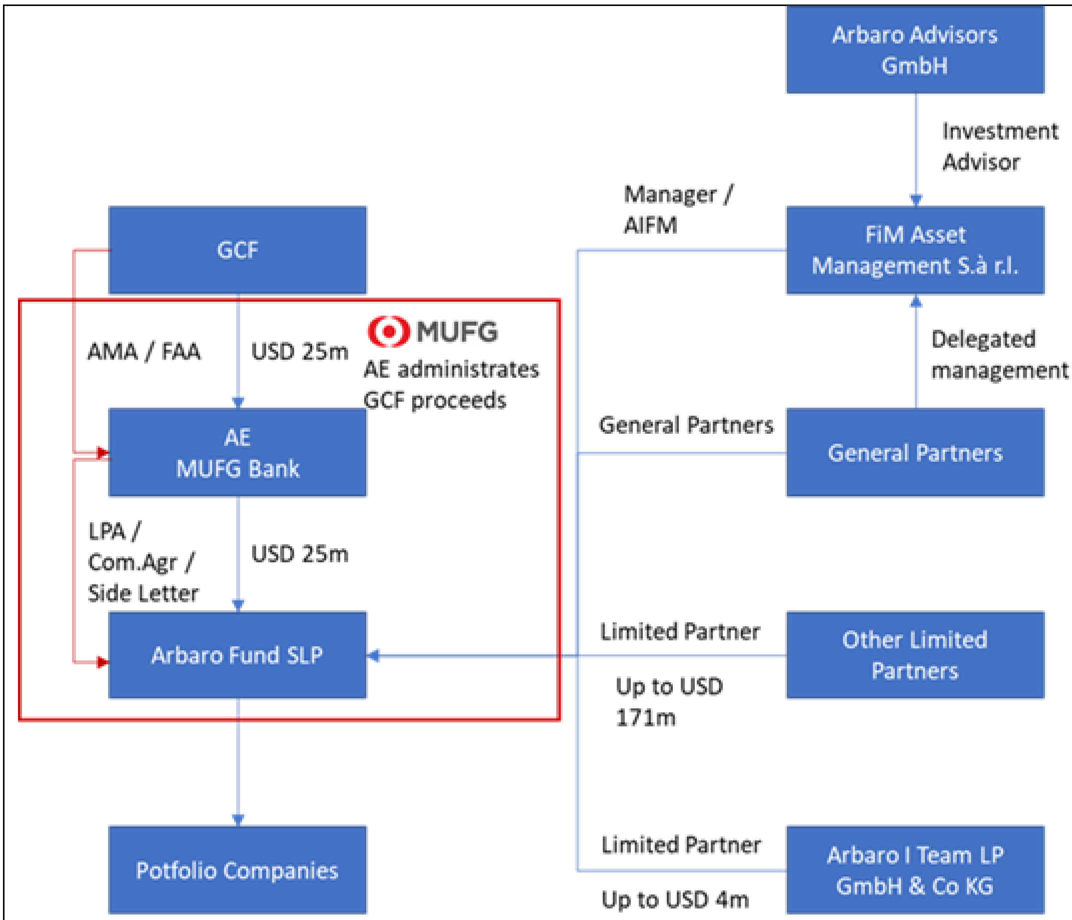
B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

<u>Description of Implementation Agreements and Funds Flow Structure</u>

MUFG Bank, as Accredited Entity will oversee the project administration, monitor the project implementation, and will insure project compliance with MUFG Bank's own relevant policies. MUFG Bank will execute an agreement with Executing Entities, the AIFM and the Arbaro Fund, to make sure that Executing Entities will be responsible for the obligations which the Accredited Entity has under the Accredited Master Agreement.

The GCF investment into the Arbaro Fund is and equity investment and the Arbaro Fund in turn will deploy funding in the form of equity or equity-like instruments (such as e.g. mezzanine or shareholder loans) to its investee projects (portfolio companies).

Figure 5 -Flow of funds is described in the graph



MUFG Bank will review the execution of the projects, monitor the capability and performance of the Executing Entities, and assess any change in circumstances that may have a bearing on the plan in general and on the implementation and operation of the projects in particular.

More specifically, MUFG Bank will:

- conduct review missions,
- monitor the Executing Entities' compliance with equity effectiveness conditions,
- monitor the procurement of goods, works, and consulting services,
- monitor implementation and development performance,
- analyse progress reports,
- disburse GCF's equity proceeds, and monitoring project cash flows,
- review unaudited and audited project accounts and agency financial statements,
- monitor the Executing Entities' compliance with applicable MUFG Bank's policies as set out in relevant agreements,
- monitor the project compliance with environmental and social safeguards, social dimensions and gender development,
- monitor Executing Entities' compliance with covenants,
- strengthening the Executing Entities' financial management and developing their capacity,
- prepare project completion reports, and
- assess the achievement of the project outcome and outputs, and the contribution to achieving the development impact.

<u>Description of Grievance Mechanism</u>

Firstly, MUFG Bank has set means to receive claims by phone calls and emails which are set out in the webpages below.

https://www.mufgamericas.com/contact-us

http://www.bk.mufg.jp/voice/index.html

Also, Japanese Bankers Association ("JBA"), which MUFG Bank is a member of, provides alternative dispute resolution ("ADR"). Japanese Bankers Association concluded the agreement on "Strengthening its support for resolving complaints/disputes." This agreement was made in conjunction with the Mediation Committee, the support organization for dispute resolution. The aim is to strengthen support for resolving complaints/disputes and facilitate use of the support system for dispute resolution. The following is an outline of the agreement:

1. Strengthening financial alternative dispute resolution (ADR) is a means to enhance customer confidence in banks. The banking industry has taken the step to provide a fair and neutral dispute resolution vehicle that can provide prompt and transparent dispute resolution and has established measures to ensure its effectiveness for customers. Member banks are committed to endeavouring to listen earnestly to the voices of customers and to prevent troubles from occurring.
2. Member banks engaged in transactions of derivatives and specified deposits, etc. shall agree to becoming the "target business operators" of JBA as a certified investor protection organization, unless reasonable reasons for not doing so exist.
3. When the resolution of a complaint forwarded to each bank or consumer relations office appears difficult, the member bank shall refer the subject to the Mediation Committee with the consent of the customer, and endeavour to promptly resolve the complaints in accordance with the mediation proposals. When a customer wants to utilize the Mediation Committee, etc., priority should be given to using the support organization for the dispute resolution that the customer wants to make use of.
4. Member banks shall observe the "Rules for Facilitating Complaint Resolution and Mediation," and sincerely handle the claims to resolve them smoothly.
5. JBA shall proactively conduct public relations by means of leaflets and posters concerning consumer relations offices and the Mediation Committee.

In addition to above, MUFG Bank set out a dedicated page on MUFG Bank's website to receive comments, feedbacks or complains for GCF-funded projects. The proposed grievance mechanism has been agreed by GCF Accreditation Panel and is implemented on MUFG Bank's webpage. As for any complain for MUFG Bank and/or Executing Entities, before pursuing legal action, MUFG Bank's GCF Focal Point will receive dispute resolution requests by emails or regular mails. Please see MUFG Bank's Procurement Guidelines draft for details.

(Please visit : https://www.bk.mufg.jp/global/productsandservices/corpandinvest/gcf.html)

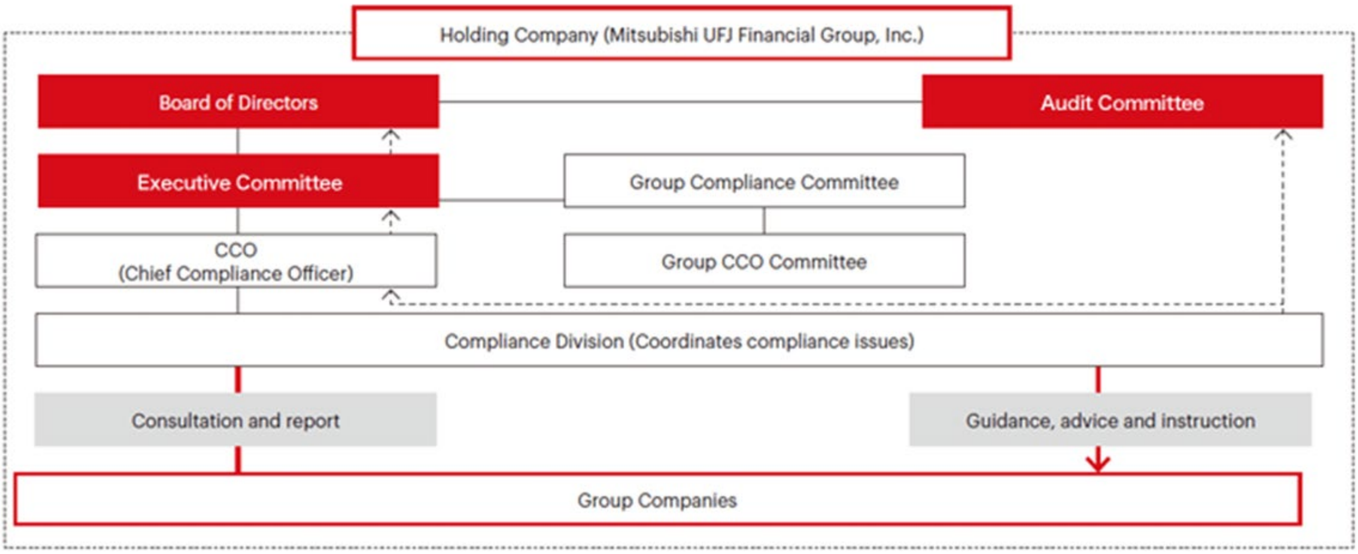
Description of MUFG Bank's Compliance Policy

MUFG Bank has clarified our group mission, long-term vision and shared values in the Corporate Vision and expressed our commitment to meeting the expectations of customers and society as a whole. Furthermore, we have established the Principles of Ethics and Conduct as the guidelines for how the group's directors and employees act to realize the Corporate Vision. This expresses our commitment to complying with laws and regulations globally, to acting with honesty and integrity, and to behaving in a manner that supports and strengthens the trust and confidence of society.

In addition, as MUFG Bank expands our business globally, we are committed to keeping abreast with developments in the laws and regulations of the jurisdictions in which we operate, including those targeting money laundering and bribery, as well as competition laws, while paying attention to trends in financial crimes.

Compliance management divisions have been established at the holding company Mitsubishi UFJ Financial Group, and at MUFG Bank, Mitsubishi UFJ Trust and Banking, and Mitsubishi UFJ Securities Holdings. Each compliance management division formulates compliance programs and organizes training courses to promote compliance, and regularly reports to each company's board of directors and Executive Committee on the status of compliance activities. The holding company has the Group Compliance Committee while the three companies have Compliance Committees which deliberate important matters related to compliance. Additionally, the holding company has the Group Chief Compliance Officer (CCO) Committee composed of the CCO of the holding company and CCOs of the three companies. This committee deliberates important matters related to compliance and compliance-related issues for which the Group should share a common understanding.

Compliance System



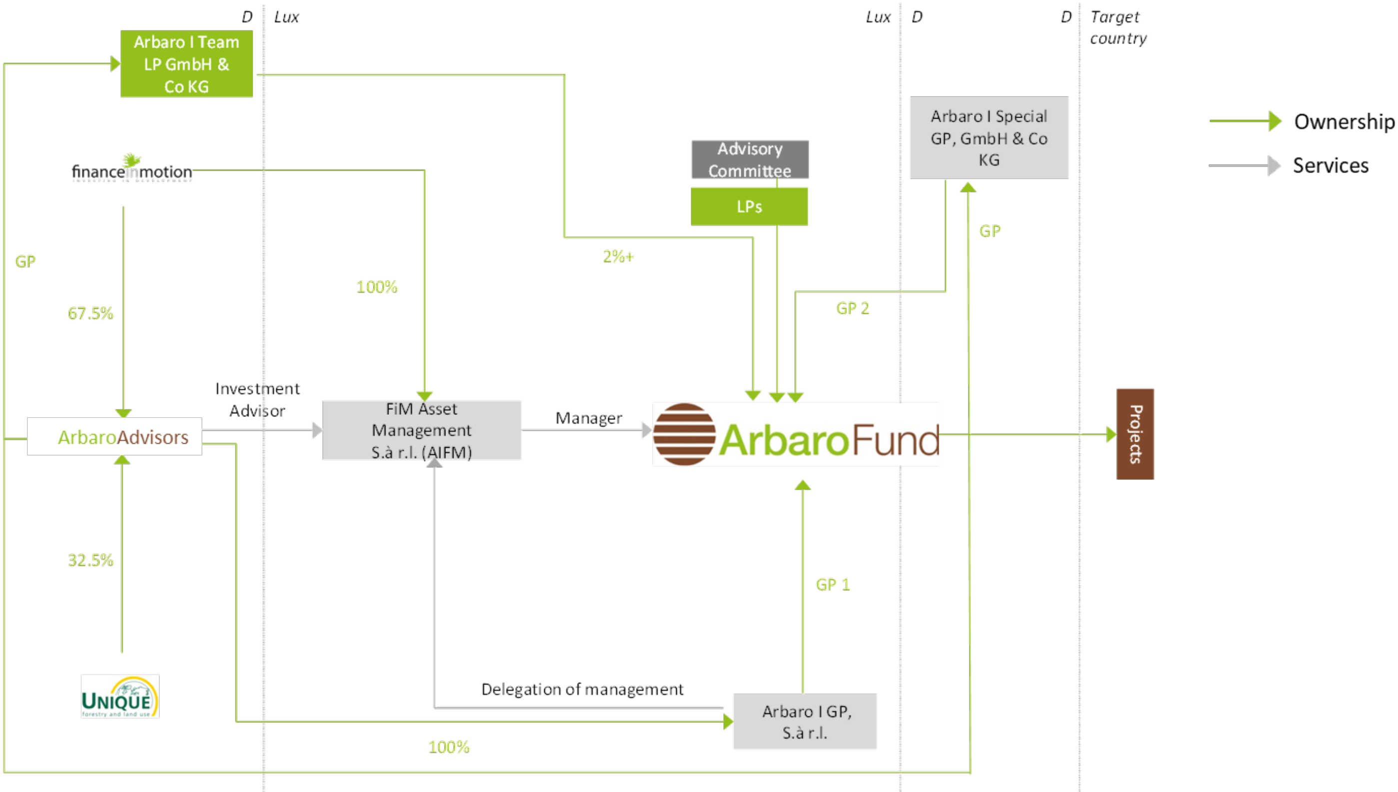
<u>MUFG Bank's Anti-Money Laundering and Counter-Terrorist Financing (AML/CTF) Program</u>

To be included in Confidential version

Arbaro Fund

Arbaro Fund, SLP is established as a Luxembourg special limited partnership (société en commandite spéciale) among the General Partner (Arbaro I GP S.à r.l), the Special General Partner (Arbaro I Special GP GmbH & co KG), and the investors as limited partners.

Please find below the structure for Arbaro Fund and its governing entities.



Arbaro Fund functions as a traditional private equity investment fund.

Arbaro Advisors GmbH acts as the Investment Advisor to the AIFM/Arbaro Fund and employs the investment team which is dedicated to Arbaro. The sole purpose of the company is advising the AIFM/Arbaro Fund on investment origination, evaluation, execution and monitoring. An Investment Advisor Agreement entered into between Arbaro Advisors and AIFM / Arbaro Fund governs this relationship. AIFM (the Manager), the General Partner and the Arbaro Fund SCS are the Executing Entities of the programme.

The General Partner (GP) is owned by the Investment Advisor and has no other purpose than acting as the General Partner for the Arbaro Fund. The GP appoints the AIFM and the Investment Advisor.

FiM Asset Management S.à r.l. acts as the AIFM of the Arbaro Fund. As the manager of the Arbaro Fund, the AIFM among other things will review the investment proposal of the Investment Advisor. The management of the Arbaro Fund shall be vested exclusively in the AIFM. The AIFM is regulated by the Alternative Investment Fund Manager Directive of the EU. The AIFM is 100% owned by Finance in Motion GmbH.

Finance in Motion GmbH and UNIQUE Forest Investment GmbH are the shareholders of Arbaro Advisors GmbH. Their relationship is governed by a shareholders' agreement. Finance in Motion hosts Arbaro Advisors in its offices and provide back office support on two levels: (1) on the company level it will provide accounting, HR, IT and communication support and (2) on the advisory level it will give transaction support, particularly in terms of legal advice, FX management, country and sovereign analysis. UNIQUE will provide technical back office support to Arbaro in the area of FSC certification, forest inventory, geographic information system (GIS), market intelligence, as well as environmental and social governance.

Finance in Motion GmbH as owner of the AIFM is supervised by the German financial sector regulator BaFin.

Each investor will enter into the Limited Partnership Agreement (LPA) with the AIFM / Arbaro Fund which governs the relationship and the rights and obligations of each party.

The Limited Partners shall have no part in the conduct of business of the Arbaro Fund or in the management or control of the Fund and shall have no authority or right to act on behalf of the Arbaro Fund.

Special General Partner (Arbaro I Special GP GmbH & co KG) is the entity receiving the carried interest.

Arbaro I Team LP GmbH & Co KG, is a German partnership created to channel the investment of the Investment Advisor (partners, managers, team members or other employees of the Investment Advisor, the Manager or any of their Affiliates and the members of the Investment Committee) in the Arbaro Fund as Limited Partners. Its commitment shall be equal to the higher of (i) two million US Dollars (USD 2,000,000) or (ii) in aggregate at least 2% of the Total Commitments as from the Initial Closing Date (the "Advisor Commitment").

Advisory Committee of the Arbaro Fund
The advisory committee of Arbaro is composed of representatives of the limited partners (i.e. investors). The LPA outlines the functions of the Advisory Committee and the procedures to be followed.

Auditor

The Arbaro Fund is audited by EY Luxembourg.

Project level

Arbaro only invests in projects that meet all the investment requirements of the Arbaro Fund as defined in the legal documentation of the Arbaro Fund. The most important criteria include:

- Investment strategy, guidelines and restrictions (please refer to Annex 26)
- Environmental and social safeguards, as defined in the ESMS of the Arbaro Fund (please refer to Annex 6)
- Climate eligibility criteria (please refer to Annex 25)
- FSC certification (please refer to Annex 23)

Arbaro projects are chosen as part of a stringent investment process that consists of five steps, from early project identification to investment completion (corresponding to activities 1 and 2 presented in section B3). The investment process can be illustrated as follows:

	1 Eligibility	2 Screening	3 Pre-due Diligence	4 Due Diligence	5 Execution
Description	• Identification of potential projects • Development of long list • Benchmarking against financial and ESG KPIs and market data • Climate eligibility	• From long-list to short-list, stringent filtering of focus projects • Desk review of available information, including ESG performance	• DD by management team • Technical and organizational DD • ESG DD	• Confirmatory DD with external support (inventory, legal, tax, etc.)	• Completion of investments • Legal and ESG documentation
Decision	• Perform detailed screening	• Commence pre-DD	• Approve preliminary terms and conditions of investment • Commence DD	• Approval of investment	---

Projects identified by the team as potential investments must successfully complete all the phases of the investment process: as a result only the few projects that comply will all Arbaro requirements will become investees. The investment process involves multiple assessments, that become more detailed as the process moves forward, and requires approval of the Investment committee of the Arbaro Fund. This investment committee consists of four external members who each have expertise in fields important to Arbaro:forestry, developing countries, impact investments, private equity. At the end of the process and prior to an investment decision, a project will have been subject to detailed due diligence of all relevant dimensions: climate, environmental, social, technical, economic, financial, organisational and legal aspects. The investment process is supported by detailed guidance and assessment templates. Please refer to Annex 6 for detailed information.

At the end of the investment process, in the execution phase, the investment terms are negotiated between Arbaro and the investee. Foreseen operations, Arbaro requirements and standards, as well as any necessary measures to address weaknesses identified in the due diligence process, are included in the investment documentation. Adherence to these is part of the legal obligations of the investee, and penalties will apply in cases of non-compliance.
Once Arbaro is invested in a project, the investee starts the implementation of operations as foreseen in the investment documentation. In this phase and over the lifetime of the investment, Arbaro provides oversight and controls the investees (corresponding to activities 3 to 5 in section B3). Main activities include monitoring and reporting of the company performance and compliance with contractual commitments, active support and backstopping for strategic planning and in face of challenges, and exercising voting rights (please refer to section E7).

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

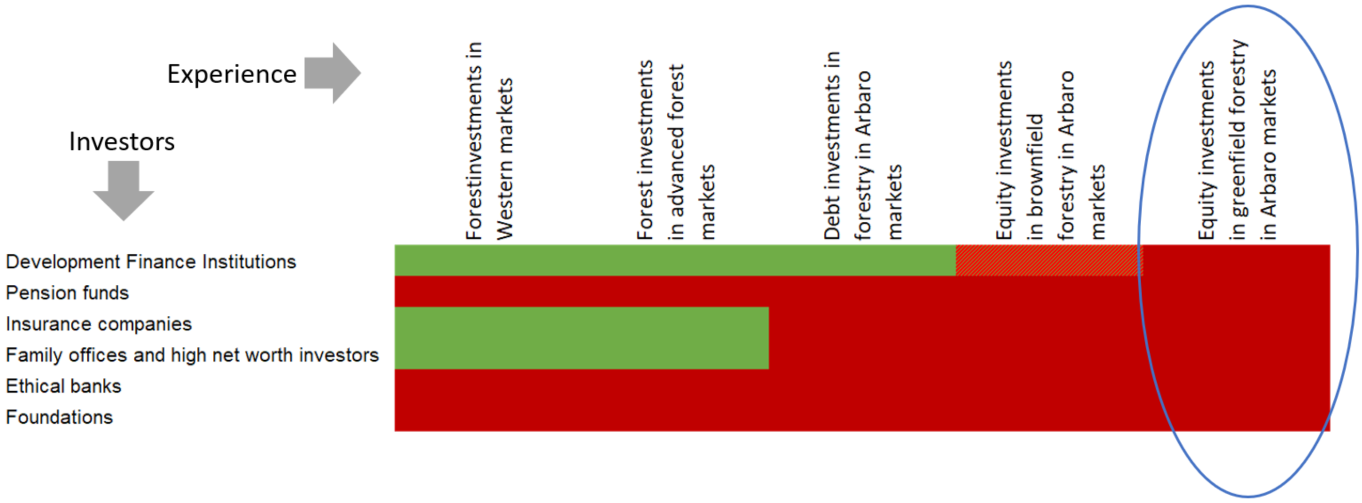
New asset class in emerging markets with limited access to finance

As an asset class, forestry investments in emerging markets and developing countries is unknown to most investors and particularly private investors have so far not been very active in the sector. DFIs/IFIs are typically among the frontrunners for new asset classes in less developed markets and can, as anchor investors, demonstrate confidence in new asset classes. However, in the case of plantation investments even the DFIs involvement has been limited so far and mainly debt has been provided.

In Arbaro's target countries there have been limited international investments in sustainable forest plantations despite the strong needs to increase the supply to the market. During Arbaro's development phase several other African forestry initiatives have failed to successfully complete their fundraising with private investors. GCF has a vital role in bringing scale to the Arbaro Fund and thus multiplying the impact that Arbaro is seeking to deliver. In addition to the funding provided and the associated impact delivered, a larger fund size will secure long term feasibility of the Arbaro Fund management. An investment by GCF, with a matching size to that of Arbaro's anchor investor would unlock the whole amount of the anchor investor and is expected to bring the commitments beyond 120 million US dollars, which is the size required to sustain the team needed to deliver the intended investment strategy and impact.

Arbaro is innovative as it will be able to mobilize new types of investors into the sustainable plantation forestry sector in the target countries. This is important in order for the sector to attract funding in scale. An investment by GCF will play a major role in mobilizing private sector capital into the Arbaro and ensuring Arbaro will reach significant scale.

The following chart shows the average experience level in forest investment of different types of investors:



Diversified projects to combat climate change

In climate change mitigation and adaptation, sustainable plantation forestry programmes have not been widely used, with Renewable energy and REDD and REDD + programmes having taken a more significant role and receiving more attention from both private and public investors. However sustainable plantation forestry when implemented with high environmental and social standards, including careful land tenure considerations and following principles of true participation, provide significant mitigation and adaptation benefits. In addition, these programmes contribute to the overall shift towards truly sustainable production of wood and wood-based materials with significantly lower land requirements than natural forest utilisation. GCF's involvement in this sector is critical in bringing it to the forefront of battle against climate change. GCF's involvement will also provide Arbaro with further credibility, which is likely to accelerate the fundraising efforts. A GCF contribution is considered a stamp of approval and ensures adherence to highest ecological and social standards as well as clear climate targets. From the fund-raising activities with private institutional investors GCFs participation in Arbaro will significantly improve the positioning of the Arbaro, particularly among private investors who do not yet have a history in climate and land use investing and hence rely on GCF's and the AE's experience.

Barriers of entry

Sustainable plantation forestry achieves its highest mitigation and adaptation impact when new forest plantations are established, e.g. on degraded land. However, the cash flow profile of such greenfield investments creates barriers to significant investment in the sector. Greenfield plantations require large upfront investments (ca. 3 000-4 000 USD/ha), but only start generating positive cash flows after several years, sometimes decades after the first investment. Even fast-growing species such as acacia or eucalyptus will need 10-12 years under favorable growth conditions to reach a sufficient height and diameter for high value end uses.

This stretched-out cash flow profile – especially of long (wood growth) rotation projects with a high mitigation potential – puts the financing structure of a greenfield project under strain, as not only initial establishment costs but also maintenance costs in the early years of operation have to be financed up front until operational costs can be covered by cash inflows. Hence, greenfield plantation projects are not bankable for conventional debt financing and thus in dire need for flexible and patient equity financing, such as the equity contributed by Arbaro.

Crowding in / Example effect

Plantation forestry in many of the countries in Arbaro's target regions has seen only little investment to date. Despite offering attractive risk-adjusted returns and significant impact potential, especially private investors have largely shunned plantation forestry in the target regions for different reasons, incl. the sector being at a nascent development stage and high perceived operational or country risks. The proposed financing structure foresees GCF to invest following the same terms as other investors into the Arbaro Fund, i.e. the GCF investment will not be concessional. Nonetheless, it is expected that GCF's investment will catalyse further co-investments into the Arbaro Fund and into plantation forestry in the target regions over the medium term. This is because the GCF investment will help Arbaro to realize more projects in more countries and thereby create additional track record to attract future investors. Similarly, the GCF investment will support Arbaro in reaching a critical scale to ensure an efficient fund structure and sufficient project diversification. As the Arbaro Fund foresees to reach an ultimate size of USD 200 million, there is plenty of room for further co-investments, meaning GCF's investment is not expected to crowd out, but rather to encourage other private and public investors to follow suit.

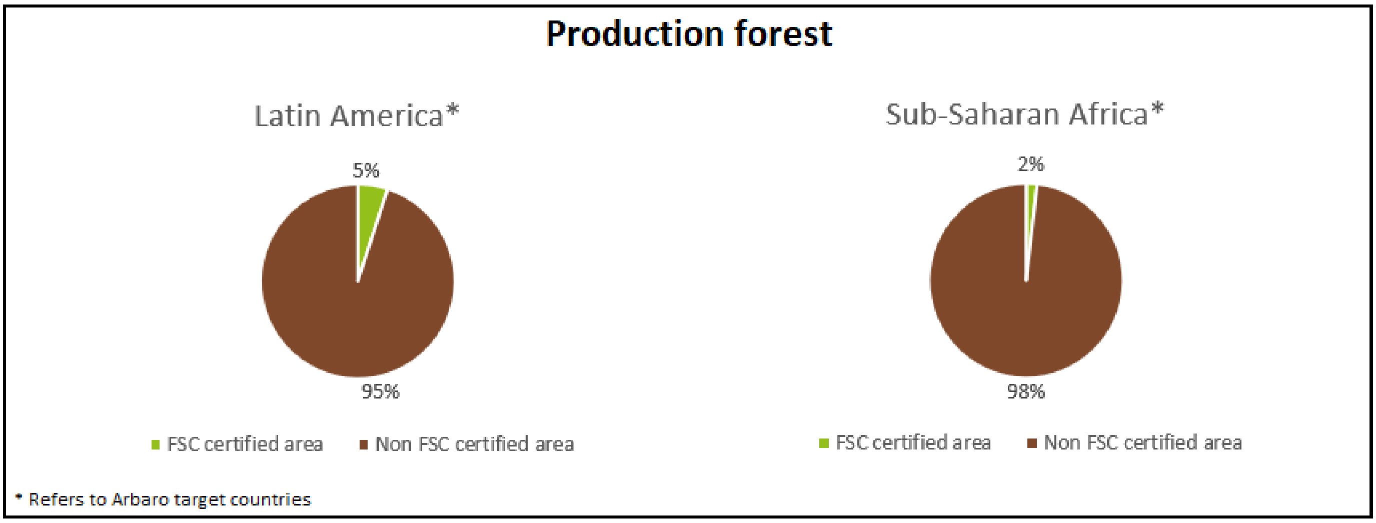
Multiplying effect of GCF investment

A GCF investment of USD 25 million would contribute to 12.5% of the ultimate fund size of USD 200 million targeted for Arbaro. Hence, every USD invested by GCF is expected to catalyse seven USD of further capital mobilised from other private and DFI investors directly into the Arbaro Fund. In addition, further capital can be mobilized indirectly on project level. Arbaro will typically not be the sole investor in the investee forestry companies, but provide financing alongside equity co-investors, such as the project developers, high net worth individuals or shareholders of the Arbaro Fund seeking direct exposure to certain projects, and mezzanine investors such as DFIs. On average, Arbaro assumes to contribute 50% of the funding needs of a project; 20-35% are expected to be financed with debt or mezzanine and the rest to be provided by other equity co-investors. Hence, GCF's investment into Arbaro is likely to be multiplied by a factor of 15x, potentially more depending on the financing structure of Arbaro's investments. Finally, Arbaro would be one of the first private equity funds to solely invest into sustainable forestry in the target regions and thus has the potential to spark follow-on investments by creating more track record for sustainable forestry as an asset class and by reducing perceived risks.

Technical innovation through FSC

In Arbaro target countries, the forest area with FSC certification represents a very small share of production forests: 5% in Latin America and 2% in Africa. Forest production under FSC standards is not a common practice in Arbaro countries, but an innovative approach that needs substantial support to become the industry standard and contribute to the shift towards sustainable production in the sector.

Figure 6 – FSC certified forests Arbaro's target countries in Latin America and Sub-Saharan Africa



When restricted to forest plantation operations, there is only a small number of companies that operate with FSC certificate in Arbaro target countries. This low number even includes international companies focused on export markets (e.g. pulp and paper producers), which usually manage large areas. With its investments Arbaro will support current and new companies to achieve FSC certification, which will improve their economic, environmental and social performance, access to finance and market position.

Table 3 FSC certified forest plantations in target countries

Country	Certified area	Number of companies
Ethiopia	0 ha	0
Ghana	21 430 ha	2
Sierra Leone	6 281 ha	1
Uganda	42 784 ha	4
Ecuador	37 138 ha	5

Paraguay	30 528 ha	4
Peru	1 100 ha	1

Note: certified area includes both plantation and protection areas

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

Arbaro investors (including the GCF)

Arbaro Fund has a limited lifetime of 15 years. This can be extended by one year with the approval of Arbaro's advisory committee. An extension of this maximum lifetime beyond 16 years requires a decision by a qualified majority of the investors (including approval of the GCF). What this means is that Arbaro will gradually start exiting the projects in later years of its term and return the capital and the profit made to the investors.

Arbaro investments - General

Arbaro takes a country-by-country approach to market liquidity and exiting its investments. A credible exit strategy with multiple options is a prerequisite for Arbaro's investment since the Arbaro has a limited lifetime. Arbaro's general goal is to establish or upgrade portfolio companies and their plantations to a stage where they can be considered high quality and well-developed brownfield investments by potential buyers. The Arbaro Fund will seek a project-specific exit strategy that maximizes the IRR while ensuring the long-term sustainability of the project beyond Arbaro's involvement.

Data availability on African and Latin American deals in our target countries is limited, but high-quality plantations and processing facilities are sought after assets in our region given how scarce they are.

Arbaro exit will be different for each project and depend on the project specific conditions. For example, in projects where Arbaro's local partner has a significant stake in the project, they may wish to gradually take over the project thus providing the Arbaro an exit. However, in each case the exit considerations will be done based on continuing the project as a going concern.

1. *Economic arguments – rational investor*

   Good assets attract good buyers: a FSC certified asset is more interesting to a buyer whose own policies and regulations require certification or who otherwise has interest in sustainability than to one without such requirements / interest – our project will all be FSC certified.

   FSC certification is likely to add a premium to the selling price of the asset; a buyer who is not interested in carrying on sustainable forest management practices would lose the certification and therefore it would be irrational to pay the premium.

   Unsustainable forest management practises (in extreme case clear cut / no replanting) would only make sense if the asset can be bought very cheap and the raw material can be quickly sold at high price, however:

   a. Selling with a high price is not possible because clear cut would mean flooding the market with excess quantities of the raw material, which would depress prices
   b. A rational buyer would therefore continue to maintain and harvest slower

   Hence, a rational investor would continue sustainable forest management practices out of economic self-interest.

2. *Regulatory dimensions*

   Various licences and approvals are needed in order to operate a plantation business. Often these would include sustainability requirements and unsustainable practises could result to losing official licence to operate.

3. *Social dimension: social licence to operate – rational investor*

   The over-time developed positive relations with different stakeholders and the social licence to operate provide an economic value but also significantly reduce the risks related to the project.

Benefits include:

- stable and motivated work force
- no litigation
- no arson
- stable off-take agreements
- stable supply agreements (out-growers), etc.

A rational investor would seek to maintain these.

4. *Know Your Customer (KYC) requirements*

Arbaro follows strict KYC guidelines and always has to identify the beneficial owner the parties it deals with as well as assess the profile of its shareholders and management. For this exercise Arbaro consults negative lists which are part of the World-Check database. In addition, it shall review that no contracts are entered into or partnerships formed with, or to the benefit of, any individual or institution named on the list of sanctioned persons promulgated by the United Nations Security Council or by the Council of the EU in connection with financing of terrorism.

5. *Contractual obligations*

Arbaro can include sustainability obligations in the (exit) sales agreements, although these will be difficult to enforce if breached.

6. *Exit options*

TYPE OF INVESTOR	MOTIVES
INSTITUTIONAL INVESTORS (pension/sovereign wealth/private equity funds, insurance companies, foundations etc.)	• Steady income and portfolio stabilization • Diversification benefits • Attractive financial returns
TIMOS AND TIMBER FUNDS	• Market penetration or expansion of market share • Realize synergies • Diversification into markets/products • Dividend income
LOCAL/REGIONAL INDUSTRIAL FORESTRY COMPANIES	• Vertical integration • Secure wood supply (from sustainable sources) • Reduce operational expenses • Learning effects
MULTINATIONAL INDUSTRIAL FORESTRY COMPANIES	• Secure high-timber-growth locations for own wood supply • Geographical diversification • Strategic expansion of business portfolio
OTHER (INTERNATIONAL) INDUSTRIAL COMPANIES	• Secure access to resources • Geographical diversification • Expand business portfolio (industry diversification) • Market penetration

Note: Other potential exit routes include sale to management/other shareholders, sale to IFIs/DFIs or impact investors, as well as IPOs

7. *Mitigation of non-permanence risks*

The reversal of achieved carbon sequestration is a risk inherent to afforestation / reforestation operations. This can happen due to losses of achieved stocks (e.g. storms or fires) or due to the interruption of management practices (e.g. when the production cycle is not completed, or harvested trees are not replanted afterwards).

To address this risk in an effective manner, Arbaro follows an approach similar to the carbon certification standards VCS and Gold Standard. Since these standards issue permanent carbon certificates, addressing the risks of non-permanence is a crucial aspect for them.

Above all both standards give great importance to the sustainability of projects: projects that are economically viable, socially beneficial and environmentally sound can continue operations in the long term. In addition, both standards consider that FSC certification ensures the sustainability of the project and a strong commitment to continue operations in the long run. This view is shared by Arbaro.

Based on a tool developed by VCS to assess non-permanence risks of projects, the following table provides an overview of non-permanence risk factors and a description of how Arbaro implements mitigation measures for all of them.

Table 4 – Non-permanence risk factors

Non-permanence risk factors	Arbaro mitigation approach
Project management: species not proved or not adapted to the site, encroachment, unprofessional or non-local teams	• Best silvicultural practices and strict site species match • Partnership with local teams and highly professional management
Financial viability: long time until breakeven or funds needed until breakeven not secured	• Companies have reached or are close to breakeven at exit
Opportunity costs: more profitable alternative land uses	• Focus on degraded land, areas designated for forestry and areas with limited suitability for agriculture
Project longevity: poor commitment to continue management practices	• Long-term operational and financial planning • FSC certification
Land tenure: resources owned by other parties, disputes over land tenure / access to resources	• Clear land tenure • No projects on land with unresolvable disputes
Community engagement: poor consultation of local communities	• Continuous engagement with communities since project start
Governance: low governance score	• Proactive cooperation with institutions
Natural risks: high likelihood and significance of fires, pests and diseases, extreme weather, geological and other natural risks	• Prevention measures and emergency plans in place

Based on the approaches followed by recognized carbon certification standards, the investment strategy of Arbaro, its commitment to achieve FSC certification in all projects, as well as the high environmental and social standards applied, ensure that risks of non-permanence are properly mitigated in all the projects and provide sufficient assurance on the sustainability and continuation of management practices in the long run.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount		Currency	
		25 million		million USD ($)	
GCF financial instrument		**Amount**	**Tenor**	**Grace period**	**Pricing**
(i)	Senior loans	Enter amount	Enter years	Enter years	Enter %
(ii)	Subordinated loans	Enter amount	Enter years	Enter years	Enter %
(iii)	Equity	25 million			12 % equity return
(iv)	Guarantees	Enter amount	Enter years		
(v)	Reimbursable grants	Enter amount			
(vi)	Grants	Enter amount			
(vii)	Result-based payments	Enter amount			

(b) Co-financing information	Total amount	Currency
	175	million USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
European Investment Bank	Equity	25	million USD ($)	Enter years Enter years	Enter%	Options
Private Investors	Equity	100	million USD ($)	Enter years Enter years	Enter%	Options
Other development finance institutions	Equity	46	million USD ($)	Enter years Enter years	Enter%	Options
Arbaro team	Equity	4,0	million USD ($)	Enter years Enter years	Enter%	Options

(c) Total financing (c) = (a)+(b)	Amount	Currency
	200 million	million USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)

The Arbaro Fund achieved its First closing in July 2018, second / subsequent closings are expected by July 2020 with the following investors / commitments:

Investor	1st Closing Amount (USD million)	2nd Closing Amount (USD million)
DFI	15.0	10.0
Private	12.0	
DFI	10.0	
Private	7.5	
Private	6.0	6.0
Private	5.0	
Arbaro team	2.0	
Private	1.2	

Private	1.5	
New investors		
GCF		25.0
DFI		15.0
DFI		10.0
Other private investors		19.0
Total	**60.2**	**85.0**
Grand total		**145.2**

Further private sector investors / DFIs with commitments amounting to ca USD 60 million are expected to commit following GCF's commitment, and the final closing is expected to meet Arbaro's target size of USD 200 million.

Further financing will be attracted at the project level either in the form of equity from project co-investors or as mezzanine or debt. Whilst the final exact amounts of this funding vary project to project, it is expected that up to a similar amount to that of Arbaro's funding will be attracted at project level from other sources.

C.2. Financing by component

Component	Output	Indicative cost Options	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institutions
Investment into plantation establishment and operations Activity 3)	New plantations created and company operations	133.2 million	16.7 million	Equity	116.5	Equity	Arbaro Fund investors, see section C1
	Land is purchased or leased	34.8 million	4.2 million	Equity	30.6 million	Equity	Arbaro Fund investors, see section C1
Fund management, transaction, custody costs Activities 1), 2), 4), 5), 6)	Fund management fees	26.7 million	3.3 million	Equity	23.4 million	Equity	Arbaro Fund investors, see section C1
	Fund costs	5.4 million	0.8 million	Equity	4.6 million	Equity	Arbaro Fund investors, see section C1
Indicative total cost (USD)		200.1 million	25 million		175.1 million		

Please see the attached budget calculation for details. Please note that the calculations assume that in the later years portion of the fees and costs are paid from dividends and exit proceeds, with the overall amount therefore exceeding the 200 million USD.

Country per country allocation is not available at this stage as the final project locations are only decided during the implementation period (investment period of the Arbaro Fund).

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)	
C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

Arbaro's projects will apply or develop towards applying best practices. This means that the projects implement state of the art silviculture in terms of modern nursery technologies, appropriate site-species matching, and growth and yield oriented thinning and pruning regimes. Particular consideration is given to work health and safety as forestry is exposed to a relatively high risk of working accidents. To ensure highest social and environmental standards Arbaro's investments will be guided by FSC Principles and Criteria and the IFC Performance Standards.

Arbaro's investments contribute to learning at individual and organizational level. The different levels of our knowledge management and capacity building is illustrated in the graph below.

Training of employees and service providers: Arbaro invests in pioneer markets for plantation forestry. Therefore, intensive training of our employees at different levels and of our service providers is a must and a permanent challenge. The training takes place in formalized courses and on-the-job.

Research & Development: Forestry is a R&D intensive business, for example in terms of matching different tree species with local site conditions or monitoring, selection / improvement of planting materials controlling pest and diseases, or optimizing operations in terms of work safety and efficiency. Arbaro will contribute to the widening of the local and regional knowledge base of forestry, as commercial plantation forestry in several countries is either at a nascent stage or largely untested. Hence Arbaro will increase the R&D potential of local enterprises or research institutions in forestry (and potentially agriculture) in the target regions. Investees' R&D departments will cooperate closely with national and international research institutions. This helps Arbaro to ensure state of the art knowledge at company level. In return, the research institutions get prove of suitability of daily use of their concepts.

Inter-company benchmarking: Arbaro will be invested in 8 to 12 forest companies in different geographies. An inter-company benchmarking allows Arbaro to learn from the best. Exchange programs between Arbaro's invested companies are part of its interregional learning approach.

Community engagement: Through community engagement programmes, Arbaro will directly invest into rural development in various topics (health, education, gender, climate smart agriculture). Additionally, Arbaro will support outgrower schemes to promote small scale forestry as an important land use for a diversified and sustainable livelihood strategy.

Interacting with forestry sector relevant authorities: Forestry has to deal and to comply with sensitive land and environment related questions. By doing forestry following international standards and best practices, Arbaro seeks to help formalizing a sector that often suffers from informality. Arbaro works closely with all forestry relevant local and national authorities.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

Mitigation impact

Arbaro will contribute to the improvement of the management of land and forest areas in its target countries through its forest plantation projects.

CO_2 sequestration: When trees are planted on non-forested land, they capture and store additional carbon dioxide from the atmosphere. To estimate the carbon sequestered by Arbaro, the overall guidance of the Intergovernmental Panel on Climate Change (IPCC) as well as methodologies from the Verified Carbon Standard (VCS) have been considered. The carbon sequestered by Arbaro projects corresponds to the net CO_2 sequestration as follows:

The net CO_2 sequestration is estimated by applying the following equation:

$$\text{Net } CO_2 \text{ sequestration } (tCO_2) = CO_2 \text{ sequestration in the project} - \text{Baseline} - \text{Leakage} - \text{Other project emissions}$$

Using this estimation methodology (for detailed methodology please see Annex 22), Arbaro projects are estimated to sequester 20 million tCO_2 during Arbaro's lifetime (15 years).

Some of Arbaro's portfolio companies may explore opportunities in voluntary carbon markets. Should any of the companies decide to proceed with issuing carbon credits (e.g. VCS or Gold standard), such credits would be deducted from the mitigation impact in the mitigation impact reporting.

It is foreseen, that such impact would be limited; assuming 200 MUSD fund, Arbaro's projects could be able to issue up to 4 million tCO_2 in credits. This only represents a small share of our 20 million tCO_2 total direct sequestration objective. The revenues obtained by companies from the sale of credits would be reinvested in the expansion of plantations, and thus result in the sequestration of additional carbon. Potentially this could also attract other investors due to better risk / return profile, diversified revenues or investors interested in carbon projects.

To account for this, the impact of the additional sequestration would be:

- Price of 5 USD per credit > 20 MUSD of additional funds for company
- Investment of 3,000 USD to establish one hectare of new plantations > 6,666 new ha
- Long term sequestration of 200 tCO_2 per ha > 1.3 million tCO_2 of additional sequestration

Therefore, the overall impact to the assumed net sequestration effect would be -2,7 million tonnes. Given the companies would only access voluntary markets, this may be over-estimation, but best estimation we can provide right now.

This means that the overall positive net sequestration would end up at 17.3 million tCO_2.

In addition, Arbaro projects will conserve (through direct conservation effort) up to 15 000 hectares of natural forest or otherwise vulnerable landscape, e.g. wetlands or habitats of endangers species. This will provide further mitigation impact during the lifetime of the programme, depending on the degradation level and rehabilitation needs of the conserved areas.

Further on, majority of the products produced from the wood will be mechanical forest products such as panels or sawn timber for products to e.g. construction. In these products the carbon stock continues throughout the lifecycle of the product, i.e. significantly beyond harvesting of the tree (in Arbaro's climate mitigation assumption this positive impact has not been accounted for).

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Climate action through forestry is underdeveloped and the potential untapped

The adoption of the Paris Agreement during the COP21 lifted the importance of forests in climate change mitigation and adaptation on the global political agenda. The adoption document writes in Article 5:

"Parties are encouraged to take action to implement and support, including through results-based payments, the existing framework as set out in related guidance and decisions already agreed under the Convention for: policy

approaches and positive incentives for activities relating to reducing emissions from deforestation and forest degradation, and the role of conservation, sustainable management of forests and enhancement of forest carbon stocks in developing countries; and alternative policy approaches, such as joint mitigation and adaptation approaches for the integral and sustainable management of forests, while reaffirming the importance of incentivizing, as appropriate, non-carbon benefits associated with such approaches."

Global mitigation and adaptation efforts in the forest sector are still in their infancy and global climate action is centered around energy; moving to renewables or increasing efficiency. Whilst changes in the energy sector are important in reaching the Paris targets, forestry has potential to significantly exceed these effects. In forestry carbon is sequestered, i.e. negative emissions are achieved. Thereby forestry has the potential to make a major contribution to climate change mitigation. This potential is particularly powerful in regions where the biophysical growth conditions are good and adequate land with low competition is available and remaining natural forests are in jeopardy due to increased product demand. Together with the significant mitigation benefits, forestry provides significant adaptation co-benefits through improved livelihoods and hence increase resilience of the rural communities but also by contribution to improved resilience of the surrounding landscapes.

Patient capital and risk-taking capital – both public and private – is necessary to create evidence and catalyze bigger investments into the sector. The sector is yet to reach these in the emerging markets. Hence, a successful launch of the Arbaro Fund has the potential to prove the environmental, social and economic concept of commercial forestry plantations and stimulate further – urgently needed – investment into the sector. It can contribute to bringing climate change mitigation and adaptation through forestry from a broad concept on paper to scalable and replicable on-the-ground climate action.

From depleting natural forests to establishing profitable carbon sinks that reduce the vulnerability of communities

Arbaro's target regions are home to many low-income countries that are focused on achieving growth and development to reduce poverty and raise standards of living. Given the significance of natural resources in these countries, there is a danger of achieving short-term growth through over-utilizing resources and under-estimating the need of being prepared for future demand. This trend is evident in the rapid depletion of indigenous forest resources in the target regions and a lack of investment in sustainable forestry.

Ca. 24 million ha of forests have been lost in Sub-Saharan Africa (ca. 14 million ha) and Latin America (ca. 10 million ha) between 2010 and 2015 alone, making the two regions the deforestation hotspots globally. Deforestation and forest degradation is mainly driven by conversion pressure from the agricultural sector and unsustainable wood extraction. While doing so helps to spur economic growth today and meet the basic needs of current generations, continuing on such a path not only reduces options for future generations but could counter economic progress made because of the important contribution that forests make to people's livelihoods, several economic sectors and climate change.

The detrimental effects of over-utilizing natural forest resources are already felt in several countries in the target regions, e.g. Paraguay or Ghana, once had flourishing wood processing industries that are now suffering. The low level of investment in the forest sector, both in processing technology and in securing long term sustainable raw material supply is paralyzing the industry. Despite outstanding biophysical growth conditions, many of such countries have slipped from significant export proceeds from these products to importing a significant share of their forest product demand. Paraguay for example went from being a net exporter of forest products to becoming a net import in the last decade. This development is expected to continue unless the sector receives significant investment in new forest plantations, improved knowledge and technology.

In addition, these developments tend to result in the market operating largely on an informal basis which in turn means less formal employment opportunities, unsustainable production, low technical and knowledge capacity as well as lost revenue in the form of license fees, taxes and tax-like payments to the local authorities. Finally, if these developments continue, the industry will further deteriorate and after some years it may be that the local supply will completely rely on imports and the informal sector, neither of which will support the long-term sustainable development goals.

Arbaro aims to work in partnerships with the local industry players, both public and private to spur and revitalize local forest industries. By investing in sustainable forestry, Arbaro will contribute as a first mover and example setter on building an industry around a renewable and reliable sustainably managed resource.

Market innovation through forest plantations

Forest plantations have existed globally for decades but in many of the Arbaro target countries, they are not an integral part of the industry and developments have been slow and few. For example, Sierra Leone has just seen its first private investment in commercial plantation forestry a few years ago. In other countries, where more investments have already been made, e.g. in Peru, silvicultural management practices are oftentimes poor and internationally recognized sustainability criteria such as the FSC are not followed. By investing in forest plantations that will be managed to the highest economic, technical, environmental and social standards, Arbaro will contribute to innovation in the local forest industries. In addition, global timber markets have already started moving away from forest products such as sawnwood from large diameter native species logs to more engineered products (e.g. finger jointing) from smaller diameter plantation logs from species like pine and eucalyptus. By providing the necessary raw materials and actively developing markets for its products, Arbaro will contribute to the necessary innovation in developing countries in the target regions to catch up with such global trends and become firstly self-sufficient in these products but going forward also export competitive.

Removing of barriers, scalability

Arbaro will contribute to scaling up the scope and impact of plantation forestry in the target regions. Commercial plantation forestry is largely at an infant stage in many of the target countries, forest industries tend to be underdeveloped and governments are just starting to appreciate the economic and climate benefits of plantation forestry, with public frameworks (e.g. regarding forest classifications, land tenure systems, community engagement, infrastructure, taxation) having significant potential to become more conducive. All this creates barriers for early movers such as Arbaro who have to invest significant time in understanding oftentimes large opaque markets and educating different local stakeholders such as policymakers, permitting agencies, financial institutions, communities or buyers about the dynamics of plantation forestry. In particular, being able to sell plantation grown timber in local markets that are both largely informal, oftentimes supplied with illegally source timber and used to native species, takes great efforts and patience.

Investments made after Arbaro's will benefit from removal of barriers and easier entry to this sector. Similarly, as forestry becomes more significant in a country, the more interest it will attract from ambitious individuals, more knowledge will grow around it and high-quality human capital will be more widely available locally.

Replicability

Similar programmes can subsequently be established in other regions or be further scaled up with additional funding. However, it is crucial to provide the example and scale it up to a level where it can be financially, ecologically and socially sustainable.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Arbaro is highly committed to inducing sustainable development in developing countries, as evidenced by the sponsors' previous activities and track record. Every investment decision will be made based upon a triple bottom line approach that takes into account the economic, environmental and social sustainability of a project as measured against quantitative and qualitative performance criteria. Next to its mitigation potential, the sustainable development impact of every project will be an integral component of the investment decision making process. Arbaro will always seek to cooperate with local project partners (e.g. forestry companies or NGOs), whose values – as demonstrated by tangible evidence – are aligned with Arbaro.

Social co-benefits including adaptation co-benefits from project level operations

Providing alternative livelihoods

Smallholder farming is an important source of employment and income for rural people in most of Arbaro's target countries but in particular in those located in the Sub-Saharan African countries. The distinctive characteristics of rural areas make them uniquely vulnerable to the impacts of climate change because of greater dependence on agriculture and natural resources makes them highly sensitive to climate variability, extreme climate events, and climate change (IPCC). Shifting cultivation remains a dominant form of agriculture across these countries. Given these practices the rural population is especially vulnerable to climate change impacts; and for example, changing rainfall patterns (prolonged dry conditions) can have a devastating impact to the livelihoods of these people. This is further exacerbated by the low financial capacity and no or limited options for provision of income. Providing alternative livelihoods through formal employment opportunities in the forestry projects or supporting services improves the resilience of the rural communities and decreases dependence on the climate sensitive subsistence practices such as shifting agriculture.

Passing best practices, incl. employment formalization

The programme will establish and pass on best practices through its operations and thereby contribute to sustainable development. For example, a large contribution will be made through the formalization of labour, which especially in rural areas is widely unknown in several of the initial target countries. If applicable, this will allow people employed under the programme to benefit from wider social services like public social security or health insurances, which they would miss if informal employment standards continued to prevail. Similarly, best practices regarding working hours, health and safety standards, vacation, etc. will be followed, which will benefit people employed under the programme beyond the monetary value of their salaries.

Contributing to developing local communities (incl. infrastructure, health and education)

Every project realised under the programme will have a community development/engagement component where investments are made into e.g. local infrastructure (e.g. wells, roads or bridges), health services (e.g. community clinics, nurses, malaria/HIV prevention) or education (e.g. school buildings, learning materials or scholarships) for the direct betterment of the livelihoods of local communities beyond the other wider benefits described in the sections above. Regarding infrastructure, it needs to be noted that oftentimes communities will also directly benefit from infrastructure developed by the project companies for commercial purposes such the expansion and improvement of road networks or building of bridges.

Reducing the vulnerability of local communities through improved agricultural practices

Rural areas in developing countries are characterized by dependence on agriculture and natural resources; high prevalence of poverty; isolation and marginality; neglect by policymakers and lower human development. The distinctive characteristics of rural areas make them uniquely vulnerable to the impacts of climate change and existing vulnerabilities caused by poverty, lower levels of education and isolation can all aggravate climate change impacts in many ways. The livelihoods of rural communities in the target regions of the programme rely on subsistence agriculture with no or only limited alternative employment opportunities, making them particularly vulnerable. Projects realised under the programme might involve outgrower schemes or community development programmes that will improve agricultural practices and hence reduce the vulnerability of local communities to climate change. Outgrower schemes will allow local farmers to use their land more efficiently and diversify their income base. In addition, certain tree species can be more resilient to climatic fluctuations that certain agricultural crops. Outgrower schemes will support such farmers to select crops that are suitable for their land, thereby reducing climatic risks. Similar outcomes can be achieved through community development programmes (see above) that improve agricultural practices e.g. through education or the provision of improved planting materials.

Mitigating the causes of urbanization

By establishing forest plantations in the countryside, the programme will generate labour in rural areas with no or only limited alternative employment opportunities – with subsistence agriculture oftentimes being the only form of economic activity. Doing so, the programme will help to mitigate the causes of urbanization, as particularly young people tend to migrate to cities to find employment and overcome economic disparities. Retaining such talent in rural areas will support the betterment of local communities and their well-being in the long run.

Gender-sensitive development impact

The projects realised under the programme will seek to offer employment opportunities or other benefits following a gender responsive approach. This means that Arbaro will use its best endeavours to ensure that the social and economic benefits that its stakeholder groups derive from the operations of its portfolio companies will be distributed in an equal basis taking into account the needs of the various groups and noting that such needs may vary between the groups.

From the sponsoring entities' previous experience, it is clear that forestry projects generally have certain areas of operation which are well suited and liked by female employees. Female planting teams have been successful in other projects for example in East Africa. Also, nursery operations typically attract female employees. It is therefore estimated that between 15 -30 % of the workforce will be female. This is seen as both reducing gender inequalities in the target regions and the vulnerability of local communities to climate change. Firstly, women living in rural areas of the target regions are oftentimes left with no employment opportunities other than – at most – subsistence agriculture, charcoal production or gathering non-wood forest products. Secondly, providing employment to women diversifies the income streams of families away from subsistence agriculture and employment of working men and thus reduces the vulnerability of their livelihoods.

Environmental co-benefits

Conserving natural forests and other natural ecosystems

The design and layout of FSC certified plantations is required to protect and conserve natural ecosystems, particularly natural forests. Wildlife corridors, streamside zones and a mosaic of stands of different ages and rotation periods must be taken into account in the layout of the plantation. FSC requires that in each certified operation multi-stakeholder processes take place to identify high conservation value areas. Hence, when developed and managed in a sustainable manner and following best practises, a plantation project will always in itself include an element of direct conservation. FSC requires that at least 10% of the project area is set aside for conservation. However, in Arbaro's projects, on average 20% of the total land area of a project will be conserved, in an active manner, i.e. such that the project is truly responsible for it. Through these practises, Arbaro envisages, through its projects to actively conserve approximately 200 ha for every 1 000 ha planted. Hence the overall direct and active conservation target of the Arbaro Fund is 15 000 ha, assuming a total investment volume of USD 200 million and depending on the final target project selection..

Enhancing degraded landscapes and improving resilience

All projects realised under the programme will seek to become FSC certified and hence follow strict environmental requirements. The environmental impact of forest plantations depends on the baseline. The programme will under no circumstances support the conversion of natural forests into forest plantations. In contrast, the programme will seek to afforest or reforested degraded landscapes for the commercial, yet sustainable production of timber resources. Besides creating additional carbon sinks, establishing forest plantations on degraded land helps to regulate soils and water flows. For example, by preventing soil erosion and water run offs, forest plantations contribute to halt the detrimental effects of land degradation. Furthermore, when forest plantations are established in between mosaics of still existing natural forests (e.g. in areas where deforestation is still on-going, but did not yet fully unfold) coupled with conservation efforts, they can provide migratory corridors to insects and other animals, hence contributing to preserving the biodiversity in such areas.

Reducing pressure on natural forests

One of the key drivers of natural forest degradation and ultimately deforestation in the target regions, especially in Sub-Saharan Africa, is informal and unsustainable timber extraction for multiple end uses such as charcoal, sawnwood or plywood production – oftentimes driven the high demand of a growing domestic population. By generating a sustainable timber resource, the programme will provide an alternative to natural forests as a source of raw material and thereby indirectly contribute to wider conservation efforts by reducing extractive pressures on natural forests over time.

Raising environmental awareness among local communities

Each project realized under the programme will actively raise the environmental awareness of local communities partly through operational risk mitigation practices and partly through community development activities. For example, raising awareness among local communities about the causes and detrimental effects of fires, landscape degradation through anthropogenic forest fires will be reduced. Similarly, the planting and conservation efforts of the different projects will raise the appreciation of local communities for the environmental value of forests as such and transfer knowledge of best practices in efficiently and sustainably utilising land. For example, outgrower schemes and community development programmes – if applicable – will educate subsistence farmers about improved agricultural practices and support them in increasing the biodiversity on their land and improve in selecting fit species for their sites.

Economic co-benefits

Generating employment

Arbaro's projects are estimated to provide direct employment to more than 5 000 people. In addition, it is assumed that each direct job will create another 2-4 jobs along the value chain, i.e. that another 10 000-20 000 jobs will be created indirectly. Further on, it is estimated that 8-10 people per each employee are supported as part of employee's extended family or through increased economic affluence and activity. This results in improved livelihoods for 200 000 people in rural communities. Importantly, generating employment especially in rural areas is key to providing alternative livelihoods to subsistence agriculture and thus to reducing the vulnerability of local communities to climate change.

Widening the local tax base

The programme will widen the local tax base in several ways. Among many other things, the plantation investments will generate profits, create employment, and fuel further investment in the resource starved processing sector. The forest sector in many of the initial target countries is dwindling due to a shortage in raw materials and the forest sector tends to be highly informal, i.e. taxes, both at individual employee level and at company level, as well as other payments such as licence fees, royalties etc. are not paid. This informality also leads to low participation in social welfare structures such as healthcare plans and pension schemes. Further growth and formalization in the forestry industry will create significant tax income for local governments, both directly through corporate income taxes and tax-like payments such as licensing fees or harvesting royalties paid by the project companies and indirectly through personal income taxes of employees and payments of other companies that benefit from the programme indirectly (e.g. contracted service providers or processing companies). This increase in tax and tax-like revenues for local governments has the potential to free up public finances that can be used elsewhere to spur sustainable economic development or climate action. While the positive impact on public finances is definite, particularly spillover effects are difficult to quantify. The below table gives an indication of the potential improvements of public finances that the Arbaro Fund is expected to induce based on broad simplified average assumptions for both target regions.

Table 5 – Approximate tax revenues

Approximate tax revenues	**USD million per annum**
Corporate income taxes - plantations	3
Corporate income taxes - processors	4
Corporate income taxes - contractors	0.2
Individual income taxes - direct employment	7
Individual income taxes - indirect employment	6
Total	**20**

In total, the Arbaro Fund is foreseen to generate some USD 20 million of annual tax income for local governments once the plantations mature. The biggest contribution will be made through individual income taxes paid by direct employees of the project companies (ca. USD 7 million per annum), followed by individual income taxes from employment generated indirectly along the value chain (ca. USD 6 million per annum). For the sake of estimating tax revenues for the government, the calculations above conservatively assume one formal tax paying job to be generated indirectly per direct job in the project companies (5 000 in total), whereas the multiplier can actually be expected to be in the range of two to four (see previous parts of the proposal). While every project will take some time to reach full scale, individual income taxes can be collected relatively early, i.e. will generate immediate tax revenues for local governments. Once the plantations start harvesting trees and reach sustainable harvesting cycle (i.e. a balanced age structure), additional direct corporate income taxes of ca. USD 3 million per annum are foreseen to be generated through the project companies. In addition, the project companies are expected to hire local contractors (e.g. for planting, maintenance and harvesting operations) and supply local wood processors such as sawmills that will generate further tax revenues of ca. USD 4 million per annum. In addition, further public revenue can be expected from tax-like payments such as licensing or land lease fees and harvesting royalties as well as formalization spillover effects along the value chain, i.e. wider positive effects on the capability and willingness of individuals and companies to pay taxes in the countries of operation (e.g. through improvements in labour standards, invoicing or electronic payments). Similarly, the above estimates do not account for the increase in tax revenues for local governments from wider positive economic spillover effects of the Arbaro Fund, for example from attracting further investments into the forestry sector by contribution to market creation and a reduction in perceived country risks.

Substituting costly imports and thereby improving the current account balance

According to FAO statistics, Latin America and Sub-Saharan Africa imported forest products worth on average USD 17 billion per year from 2011-2015; forest product imports are likely to increase in the future with on-going economic and population growth in the target regions. While some countries like Brazil, Chile, Uruguay or South Africa have advanced forest sectors that generate significant forest product trade surpluses and other countries especially in Sub-Saharan Africa like Ghana or Mozambique export basic products like sawnwood or unprocessed logs on large scale e.g. to Asia, many developing countries in the target regions rely on importing several key, mostly higher value processed forest products, such as higher grade lumber, plywood, structural/engineered timber or treated utility poles, at high costs from overseas. For example, several African countries import utility poles from destinations as far as Chile, Finland or Australia due to a lack of domestic resources. Such dynamics put the already strained current account balances and local currencies of many countries in the target regions under additional pressure. By generating sustainable domestic timber resources, Arbaro will foster the development of local forest industries and thus contribute to improving the forest trade balance of such countries. Thereby it can ultimately support the betterment of their current account balances and foreign currency reserves.

Increasing economic diversification and reducing vulnerability to external shocks

Several countries in the target regions of the programme are highly reliant on a few export products and economic growth thus very sensitive to price swings in such products. For example, Paraguay's real GDP grew by 13% in 2010, followed by a slowdown to 4% in 2011 and a contraction of -1% in 2012, only to surge again to 14% in 2013. Paraguay's economy is highly dependent on soy and beef exports (almost 50% of exports) and thus vulnerable to external price shocks and extreme weather phenomena like "El Nino". This is seen in the economic contraction of 2012 that was mainly driven by country-wide drought coupled with a slump in global prices for agricultural products. By contribution to the development of a significant local forest sector, the programme can support countries like Paraguay in diversifying their economies and becoming less vulnerable to external shocks. In fact, in the case of

Paraguay, its economy used to be more diversified because the country was home to a flourishing forest sector that however relied heavily on supply from natural forests and now crumbles due to depleting natural resources.

Transferring knowledge and building capacities

Arbaro will invest in projects that will be guided by international best practices such as the FSC Principles and Criteria and the IFC Performance Standards. The projects will be managed towards economic sustainability by becoming inherently profitable and worthwhile economic ventures. At the same time, the projects will seek to utilize local labour to the highest extent possible and invest in training employees. Coupled with the on-the-job experience, this will lead to a transfer of knowledge and best practices as well as capacity building in the target regions, hence ultimately contributing to long-term economic growth and sustainable development.

Contributing to R&D potential in developing countries

Forestry is a R&D intensive business, for example in terms of matching different tree species with local site conditions or monitoring and controlling pest and diseases and development trials or programmes on native species and their suitability for plantation purposes. As an example, in Arbaro's second investment in Paraguay, an R&D trial has been commenced. This trial will establish plantation lots with native species under different silvicultural set-ups co-operating with a local research project for the production of high quality planting materials. The programme will contribute to the widening of the local and regional knowledge base of forestry, as commercial plantation forestry in several countries is either at a nascent stage or largely untested. Hence, the programme will increase the R&D potential of local enterprises or research institutions in forestry (and potentially agriculture) in the target regions.

Supporting local SMEs along the value chain

The programme is expected to have significant spillover effects to local forest industries and related economic sectors. First and foremost, the well-being and survival of local wood processing industries such as furniture highly depends on the availability of affordable raw materials, i.e. timber. Many companies in the wood processing industries of the initial target countries are small or medium sized enterprises, e.g. carpentry businesses employing a handful of workers. Such enterprises still are or used to be significant in several of the initial target countries, but are suffering from surging raw material costs due to dwindling timber resources, low quality roundwood supply and ultimately import competition as a result of increasing prices. Such dynamics with more and more enterprises going out of business can be observed in both countries that mainly rely on natural forests as source of raw material (e.g. Ghana or Paraguay) and countries where processors mainly source timber from (constrained) plantations Arbaro will contribute to reversing such trends by establishing sustainable local timber resources in the target regions.

Secondly, the projects realised under the programme – where available and suitable – will contract local service providers, e.g. for tree maintenance or transport activities. Many such contractors in the target regions are SMEs, meaning the programme is expected to spur their local business activity. Finally, if such enterprises are contracted, they will be required to ultimately abide by the same business practices like the project companies, e.g. in terms of health and safety. This way the programme will contribute to passing best practices along the value chain and create wider economic (e.g. increased productivity) or social (e.g. increased safety standards) benefits.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Countries in Arbaro's target regions are to a large extent recipients of development assistance and classified as such in the OECD DAC listing. In these countries rural population is largely dependent on subsistence agriculture and natural resources for their livelihoods; formal employment being scarce. Large proportion of households are below the poverty line, with rural poverty being much higher than urban poverty. Since large proportion of natural resources are effectively subject to open access, these resources act as safety nets to households that have little or no other means of income, and as a means of reducing risks and smoothing income by providing the option for deriving food and income from multiple sources. With population growth, increase in population density in rural areas, as well as the general overutilization of the resources, in many areas the resources around villages have been depleted so that households are finding it more difficult to access these benefits further increasing the vulnerability of these communities.

Arbaro will make a significant contribution to improving the position of the rural communities located within the its project areas primarily by providing formal employment opportunities. It is estimated that livelihoods of up to 200 000 people will be improved through the Arbaro's activities.

In addition to providing employment and source of income, the formal employment also shifts these people to the reach of healthcare and vocation education services as well as pension and social security system.

D.5. Country ownership (max. 500 words, approximately 1 page)

Table 6 Country policies

Summary		Policies in detail						
Country	Arbaro aligned with political strategy?	National development plan	National climate policy	INDC	NAMAs	NAPAs	National forest policy	National REDD+ strategy
Sub-Saharan Africa								
Ethiopia	✓	X	X	X		X	X	X
Ghana	✓	X	X	X			X	X
Sierra Leone	✓	X	(X)			X	X	
Uganda	✓	X	X	X		X	X	X
Latin America								
Ecuador	✓	X	(X)	X			(X)	X
Paraguay	✓	X	X	X			X	X
Peru	✓	X	X	X			X	X

The Arbaro Fund's mission is aligned with the political strategy of every target country. The Arbaro Fund will contribute to climate change mitigation, and provide adaptation co-benefits to climate change and to social development in the target countries. This mission is aligned with the political strategy of each target country as evidenced in various national policies, incl. national development plans, national climate policies and INDCs submitted under the Paris Agreement. When a policy refers explicitly to reforestation/afforestation or sustainable forestry as a challenge or target for climate change mitigation/adaptation or social development, the Arbaro Fund's mission is considered aligned with the policy, i.e. a "X" was assigned in the table above. If only indirect evidence exists, e.g. because a national climate policy does not yet exist, but documentation for its preparation (Sierra Leone) or only general targets for the LULUCF exist (Ecuador) do, the Arbaro Fund's mission is considered indirectly aligned with the policy, i.e. a "(X)" was assigned in the table above. Blank fields do not indicate a conflict between the Arbaro Fund's mission and the respective country policy, but either the lack of such a policy or the absence of a reference to reforestation/afforestation or sustainable forestry in that policy. No conflict between any reviewed national strategic policy and the Arbaro Fund's mission was identified.

Every target country except for Sierra Leone explicitly intends to promote sustainable forest plantations. The main quoted motives are carbon sequestration and income generation. Sierra Leone identifies "reforestation, afforestation and agroforestry" as a mitigation opportunity in its third national communication to the UNFCC and forestry is considered a key pillar of addressing vulnerabilities and building resilience in the medium-term national development plan. Several countries formulate forest plantation strategies, incl. Chile, Ecuador, Ghana and Paraguay. The INDCs of every target country except for Sierra Leone list reforestation/afforestation or sustainable forestry as important mitigation options, typically emphasizing the contribution of reforestation/afforestation to carbon sequestration. Only few target countries have already developed NAMAs or NAPAs, but the existing action plans oftentimes cover reforestation/afforestation or sustainable forestry. All countries except for Sierra Leone have drafted or passed national REDD+ strategies, all of which consider sustainable forest plantations as a strategic option. While most of the target countries link reforestation/afforestation or sustainable forestry with social or rural development, several countries consider reforestation/afforestation or sustainable forestry ways to adapt to climate change, incl. Ethiopia, Paraguay, Sierra Leone and Uganda. Please refer to annex 23 for country-specific summaries of the alignment of the Arbaro Fund's mission with the national political strategies.

<u>Capacity of AE and Executing Entities to deliver</u>

MUFG Bank (Accredited Entity)
Building on over 350 years of financial expertise, Mitsubishi UFJ Financial Group ("MUFG") has grown to become one of the largest comprehensive financial groups in the world. Today MUFG's global network spans over 50 countries and regions, and employs 150,000 financial professionals in over approximately 2,300 offices, providing customers with a deep and diverse range of services and resources, to meet their local, regional and global financial needs.

The project finance practice within MUFG provides clients access to its global lending and capital markets businesses. MUFG has been Global No.1 in MLA League Table of Project Finance International (PFI)[16] for seven years in a row from 2012. MUFG was also ranked Global No.1 for the third straight year since 2016 in Clean Energy & Energy Smart Technology under Bloomberg New Energy Finance Asset Finance Lead Arrangers League Table.

MUFG Bank plays an active role in the Project, will channel the resources to the Project as a private accredited entity and monitor the Project's compliance with the Arbaro Fund's standards (social, environmental, fiduciary and gender). MUFG Bank's supervision of the Project will be based on a periodic review of the Project provided by the Executing Entities.

Arbaro level

The Executing Entities needs to succeed in the following aspects for the Programme / Arbaro Fund to be successful:

1) Attract additional capital first to first closing and secondly to scale the Fund to USD 200 million
2) Set up the Arbaro Fund structures
3) Originate and execute the Arbaro Fund investments in a timely manner and in accordance with the fund target impact and financial returns
4) Run the investment such that they will increase in value and provide the desired impact
5) Exit the investments

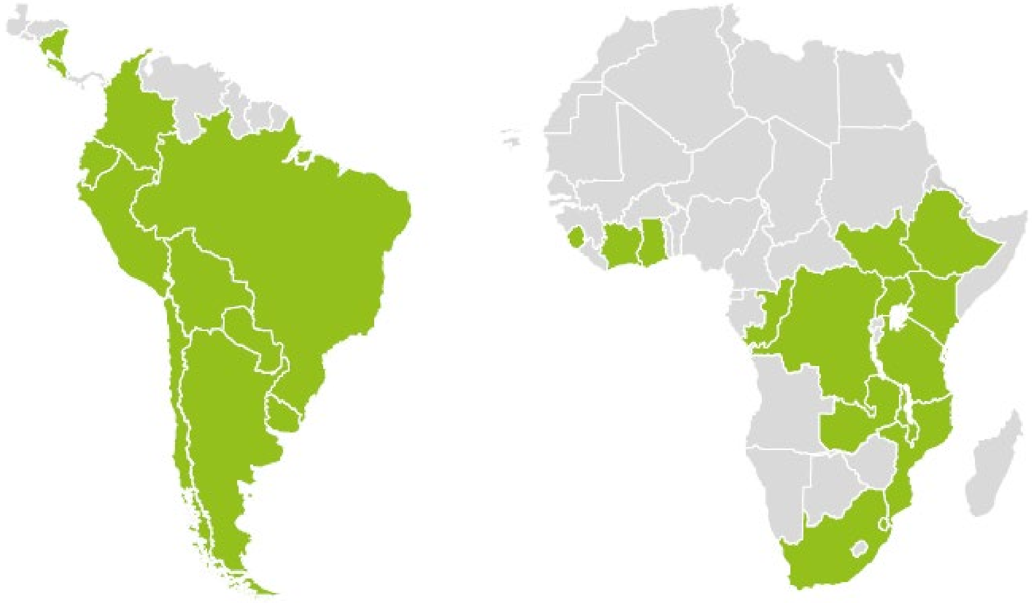
Arbaro Advisors GmbH is the Investment Advisor of the Arbaro Fund. Arbaro Advisors is owned by the Fund promoters, Finance in Motion GmbH ("FiM") and Unique Works AG ("Unique"). The Arbaro Fund promoters have a strong track record in emerging markets, working with a wide range of investors as well as in (in case of Finance in Motion) fund structuring and developments.

Arbaro Advisors will employ the investment team which will be dedicated to the Arbaro Fund. The key members of the Investment team have significant experience in the target countries and sectors as follows:

- Dr Markus Grulke, previous head of the forest investment team at Unique. Dr Grulke has experience in deal sourcing, due diligence and structuring of investments in Africa and Latin America. He has been involved in the development and co-management of different forest companies. He has also explored investment opportunities in Latin America and Africa (Paraguay, Uruguay, Chile, Peru, Nicaragua, Ecuador, Argentina, Mozambique, Tanzania, Uganda), accumulating long-standing experience in conducting forestry, economic, environmental and social due diligence missions and feasibility studies and pre-structuring of forest investments in these regions.
- Hanna Skelly has sourced, executed and managed a portfolio of approximately EUR 170 mm, focusing on forestry investments and development finance in Sub-Saharan Africa and Latin America, specifically in several of the main target countries of the Arbaro Fund (Ghana, Sierra Leone, and Uganda) among others.
- Marco Kaiser, with experience in product and strategy development and planning of FiM, overseeing FiM's Asset and Liability Management. Lead several transactions of FiM and the advised funds, with the latest investment in an African SME holding. He has broad experience in emerging countries, mainly focused on Western Balkan, South East Europe, Caucasus, Middle East, North Africa and Latin America.
- Thomas Ruf, who has developed and managed forest and timber industries in South America. He has experience in preparation of business plans for future investment projects, timber market research, valuation of forest based on natural and economic criteria, financial and legal due diligence and in forest management and production.
- Patricia del Valle, with experience in economic modelling of forestry projects, development of monitoring and forest information systems, E&S issues and assessments of forest investments in Paraguay, Argentina, Brazil, Nicaragua, Ethiopia, Mozambique, among others.
- Francisco Gomes, an investment professional with 5 years of experience in private equity deals, focusing in financial modelling, due diligence and deal execution.
- Mikko Reinikkala, a finance professional with 3 years of experience in support functions in asset management industry.

The following map highlights the countries where Arbaro team has direct experience:

[16] http://www.pfie.com/Journals/2018/01/23/m/i/k/PFILeagueTables2017.pdf



Arbaro Advisors’ owners

FiM is an investment advisor specialized in triple bottom line investments that advises development finance funds in achieving significant economic, social and environmental impact in emerging markets and developing countries. FiM is invested in 25 countries advising funds with currently outstanding assets under management of USD 1.8 bln. Cumulatively, FiM has advised more than USD 3 bln assets since its inception in 2009. The company is particularly strong and known as a leading advisor to public-private partnership funds leveraging donor contributions (e.g. BMZ, EC) and DFI investments (e.g. KfW, IFC, EBRD, EIB, FMO, OeEB) with private institutional investors. In addition to the investment funds FiM is operating technical assistance facilities for several donors with average annual approvals of ca USD 6m (400+ projects approved in last 5 years). FiM has an office network of 18 offices in three continents, including Latin America and Africa.

UNIQUE is and independent land use, timber advisory and forest management company, with experience in forest management planning, structuring forest investments, investment scoping and feasibility studies, technical, economic environmental and social due diligence of forestry projects and companies, development of forest investment projects in compliance with FSC; forest planning and operational management, process optimization and coaching of local forest company managers, market intelligence and value chain integration, forest product commercialization, forest information systems, forest valuation as well as monitoring and evaluation of productions, costs and E&S impact, FSC certification. UNIQUE has subsidiaries in Kampala (Uganda) and in Asunción (Paraguay). Also, it has regional and project offices in Uruguay, Kenia, Serbia, Vietnam, Kyrgyzstan, Ethiopia and Argentina.

Since its foundation, UNIQUE has carried out over 650 assignments in more than 75 countries.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

Adequacy of the financing structure

Sustainable plantation forestry achieves its highest mitigation and adaptation impact when new forest plantations are established, e.g. on degraded land. However, the cash flow profile of such greenfield investments creates barriers to significant investment in the sector. Greenfield plantations require large upfront investments (ca. 3 000-4 000 USD/ha), but only start generating positive cash flows several years, sometimes decades after the first investment depending on several factors. For example, if eucalyptus trees are mainly produced for biomass or pulpwood under favourable growth conditions, they can be harvested 5-8 years after planting. If, however the target is to produce higher value timber e.g. for the construction or furniture industry, trees can take more than 20 years to mature depending on species selection and growth conditions. Even fast-growing species like eucalyptus will need 10-12 years under favourable growth conditions to reach a sufficient height and diameter for high value end uses.

The stretched-out cash flow profile – especially of long rotation projects with a high mitigation potential – puts the financing structure of a greenfield project under strain, as not only initial establishment costs but also maintenance costs in the early years of operation have to be financed up front until operational costs can (at least partially) be covered by cash inflows from commercial thinnings. Hence, greenfield plantation projects are not bankable for conventional debt financing and thus in dire need for flexible and patient equity financing. In fact, domestic debt financing e.g. through commercial banks in the target regions is largely non-existent and only some DFIs have provided mezzanine debt to greenfield plantation companies – although typically with equity like features.

Plantation forestry in many of the countries in Arbaro's target regions has seen only little investment to date. Despite offering attractive risk-adjusted returns and significant impact potential, especially private investors have largely shunned plantation forestry in the target regions for different reasons, incl. the sector being at a nascent development stage and high perceived operational or country risks. The proposed financing structure foresees GCF to invest at the same terms like other investors into the Arbaro Fund, i.e. the GCF investment will not be concessional. Nonetheless, it is expected that GCF's investment will catalyse further co-investments into the Arbaro Fund and into plantation forestry in the target regions over the long run. This is because the GCF investment will help Arbaro to realize more projects in more countries and thereby create additional track record to attract future investors. Similarly, the GCF investment will support Arbaro in reaching a critical scale to ensure an efficient fund structure and sufficient project diversification. As the Arbaro Fund foresees to reach an ultimate size of USD 200 million, there is plenty of room for further co-investments, meaning GCF's investment is not expected to crowd out, but rather to encourage other private and public investors to follow suit.

Cost effectiveness and efficiency

Capital mobilisation

A GCF investment of USD 25 million would contribute to 12.5% of the ultimate fund size of USD 200 million targeted for Arbaro. Hence, every USD invested by GCF is expected to catalyse seven USD of further capital mobilised from other private and DFI investors direct into the Arbaro Fund. In addition, further capital can be mobilized indirectly on project level. Arbaro will typically not be the sole investor in the investee forestry companies, but provide financing alongside equity co-investors, such as the project developers, high net worth individuals or shareholders of the Arbaro Fund seeking direct exposure to certain projects, and mezzanine investors such as DFIs; in certain cases, Arbaro will also become a minority shareholder. On average, Arbaro assumes to contribute 50% of the funding needs of a project; 20-35% are expected to be financed with debt or mezzanine and the rest to be provided by other equity co-investors. Hence, GCF's investment into Arbaro is likely to be multiplied by a factor of 15x, potentially more depending on the financing structure of Arbaro's investments. Please refer to section E.6.2 for more details on the co-financing ratio. Finally, Arbaro would be one of the first private equity funds to solely invest into sustainable forestry in the target regions and thus has the potential to spark follow-up investments by creating more track record for sustainable forestry as an asset class and by reducing perceived risks; however, the resulting indirect capital mobilisation potential is difficult to estimate.

CO_2 sequestration

Based on a total fund size of USD 200 million, it is estimated that a total of ca. 20 million tons of CO_2 will be sequestered directly over Arbaro's lifetime. Hence, every 10 USD invested will contribute to the sequestration of approximately 1 ton of CO_2. This estimate is deemed conservative because it does not account for (i) emission reductions from indirectly preventing natural forest degradation and deforestation (see section E.3) and (ii) emission reductions after Arbaro's exit. GCF has not funded plantation forestry proposals before, meaning there are no directly comparable benchmarks. Nonetheless, the estimated mitigation costs of 10 USD/tCO_2 are competitive when compared to other – although less comparable – GCF funded projects. While landscape/REDD proposals report mitigation costs of 6-8 USD/tCO_2, renewable energy proposals project mitigation costs as low as 6 USD/tCO_2 up to of ca. 70 USD/tCO_2. Experience from other (non GCF funded) renewable energy projects in developing countries confirms that mitigation costs strongly depend on the technology and scale of a project, but can be significantly higher than what is estimated for the Arbaro Fund.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's Performance Measurement Frameworks under the Results Management Framework to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

Please select the appropriated expected result. For cross-cutting proposals, tick both.

☒ Shift to low-emission sustainable development pathways

☒ Increased climate resilient sustainable development

E.2. Core indicator targets

Provide specific numerical values for the GCF core indicators to be achieved by the project/programme. Methodologies for the calculations should be provided. This should be consistent with the information provided in section A.

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO_2 eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	1,333,333 t CO_2 eq
	Lifetime	20,000,000 t CO_2 eq
E.2.2. Estimated cost per t CO_2 eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	200 million USD
	(b) Requested GCF amount	25 million USD
	(c) Expected lifetime emission reductions	20 million t CO_2eq
	(d) Estimated cost per t CO_2eq (d = a / c)	10 USD **/ t CO_2eq**
	(e) Estimated GCF cost per t CO_2eq removed (e = b / c)	1.25 USD **/ t CO_2eq**
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	Post GCF participation (excluding 1st closing and public investors already engaged prior to GCF interest):	
	(f) Total finance leveraged	205 million USD
	(g) Public source co-financed	10 million USD
	(h) Private source finance leveraged	195 USD
	(i) Total Leverage ratio (i = f / b)	8.2
	(j) Public source co-financing ratio (j = g / b)	0.4
	(k) Private source leverage ratio (k = h / b)	7.8
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	14 000 men / 6 000 women
	Indirect	100 000 men / 100 000 women
	For a multi-country proposal, indicate the aggregate amount here and provide the data per country in annex 17.	
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	
	Indirect	
	For a multi-country proposal, leave blank and provide the data per country in annex 17.	

E.3. Fund-level impacts						
Select the appropriate impact(s) to be reported for the project/programme. Select key result areas and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected impact result. The result areas indicated in this section should match those selected in section A.4 above. Add rows as needed.						
Expected Results	**Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions**
				Mid-term	Final	
M4.0 Reduced emissions from land use, reforestation, reduced deforestation, and through sustainable forest management and conservation and enhancement of forest carbon stocks	*M4.1 Tonnes of carbon dioxide equivalent (t CO2 eq) reduced or avoided (including increased removals) - forest and land use*	Portfolio company data using agreed methodology of Arbaro	0	7 million tCO_2eq sequestered	20 million tCO_2eq sequestered.	Arbaro Fund reaches its target size of 200M USD and deploys funding in 8-12 projects in its target regions. Targets may be adjusted following guidance by the COP related to GHG emission level assessments under the NDCs. An additional indicator related to contribution to NDCs might be added following future changes in the GCF result management framework.

E.4. Fund-level outcomes						
Select the appropriate outcome(s) to be reported for the project/programme. Select key expected outcomes and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected outcome. Add rows as needed.						
Expected Outcomes	**Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions**
				Mid-term)	Final	
	Number of technologies and innovative solutions transferred or licensed to promote low-emission development as a result of Arbaro Fund support.	Portfolio company data	0	1	1	The portfolio companies use and further develop sustainable forest management practices leading to CO_2 sequestration.
M5.0 Strengthened institutional and regulatory systems	*M5.1 Institutional and regulatory systems that improve incentives for low-emission planning and development and their effective implementation*	FAO data Public data of the associations and other such institutions	Regulatory: Regulation exists but it is outdated Enforcement is lacking Sectoral: No association	Regulatory Changes in regulation Enforcement is improved but not consistent	Regulatory: New regulation is in place Enforcement is consistent	Target levels will be reassessed as part of the due diligence process of the investments.

		Portfolio company data	s exist or the have weak capacity	Sectoral: 1 new association or improved capacity of 2 associations	Sectoral 2 new associations, or improved capacity of 5 existing associations:	Qualitative indicator with two aspects: Regulatory: The forestry sector moves towards formalization and sector regulation is improved and enforced more efficiently Sector associations and other such institutions: Portfolio companies are active in sector associations and other such institutions in lobbying for more formalization.
M9.0 Improved management of land or forest areas contributing to emissions reductions	*M9.1 Hectares of land or forests under improved and effective management that contributes to CO2 emission reductions*	Portfolio company data FSC certification report FAO country data	0	25,000 ha of new forest area 5,000 ha of conservation area	75,000 ha of new forest area 15,000 ha of conservation area	Arbaro Fund reaches its target size of 200M USD and invests in 8-12 portfolio companies. Each portfolio company develops between 6,000-9,000 hectares of new forests. Each portfolio company achieves FSC certification In each project, on average 20% of the total land area is conserved. Targets are reassessed as part of the due diligence process of the investments.

E.5. Project/programme performance indicators
Arbaro will choose its investment projects during the investment period of the Arbaro Fund. As of today, the Arbaro Fund has made two investments. For details please refer to section [xx].

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Sustainable supply of wood products increases in target markets	*Volume in m3 of sustainably produced wood products entering the market or in forest stock*	Portfolio company data FSC certification reports on portfolio companies' assets	0	5 million *m3*	20 million *m3*	Arbaro Fund has a limited lifetime of 15 years. Arbaro will seek a project-specific exits that ensure the long-term sustainability of the projects beyond Arbaro's involvement. Therefore, it can be reasonably assumed that the positive impacts that the Arbaro Fund's projects provide, will continue beyond the Fund's involvement in the portfolio companies. However, the result assessment in this proposal is limited to the Arbaro Fund lifetime. Arbaro Fund reaches its target size of 200M USD and invests in 8-12 portfolio companies. Each portfolio company develops between 6,000- 9,000 hectares of new forests. Each portfolio company achieves FSC certification In each project, on average 20% of the total land area is conserved. Targets are reassessed as

						part of the due diligence process of the investments.
New forested areas are created and managed under sustainable management practices	*Number of hectares planted and sustainably managed*	Portfolio company data FSC certification reports	0	25,000 hectares	75,000 hectares	Fund size of 200M USD Each fund project will plant approximately 6,000-9,000 ha of new forest area. Each portfolio company achieves FSC certification Targets are reassessed as part of the due diligence process of the investments.
New areas are placed under conservation	*Number of hectares conserved*	Portfolio company data FSC certification reports on portfolio companies' assets	0	5,000 hectares	15,000 hectares	Fund size of 200M USD Each portfolio company develops between 6,000-9,000 hectares of new forests. Each portfolio company achieves FSC certification In each project, on average 20% of the total land area is conserved. Targets are reassessed as part of the due diligence process of the investments.
Formal jobs are created in rural areas providing alternative livelihoods to subsistence agriculture and unsustainable practices such as charcoal production	*Number of formal jobs* *Number of people with improved livelihoods*	Portfolio company data	0 0	2,500 100,000	5,000 200,000	Arbaro Fund reaches its target size of 200M USD and deploys funding in 8-12 projects. Against each formal direct job another 3 jobs are created in the value chain or

						service sectors supporting the value chain. Each employed person contributes to improved livelihoods of 8-10 persons. Increased economic activity in the rural areas provide alternative income sources. Targets are reassessed as part of the due diligence process of the investments.
New SMEs are created in the sustainable plantation forestry value chain or in supporting service industries	*Number of new entities*	Portfolio company data Baseline is assessed as part of the environmental and social due diligence	0	20	100	Arbaro Fund reaches its target size of 200M USD and deploys funding in 8-12 projects. Increased economic activity in the rural areas provide alternative income sources. Targets are reassessed as part of the due diligence process of the investments.
Local communities' capacities are enchanced through community programmes such as outgrower schemes	*Number of community programmes (e.g. outgrower schemes)* *Number of people reached by these programmes.*	Portfolio company data Portfolio company data	0 0	15 1,500	30 5,000	Arbaro Fund reaches its target size of 200M USD and deploys funding in 8-12 projects. Increased economic activity in the rural areas provide alternative income sources. Portfolio companies roll out out-grower schemes and other community programmes (e.g. training on sustainable

						agriculture) where communities in the proximity of the portfolio company operations will be trained to apply modern silviculture to their own land, thus providing several benefits such as additional and diversified income, economic empowerment and improving rural capacities. Targets are reassessed as part of the due diligence process of the investments

E.6. Activities

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in the implementation timetable. Add rows as needed.

Activity	Description	Sub-activities	Deliverables
1) Pipeline development 2) Assessment of project opportunities following Arbaro's investment strategy and ESMS	Arbaro continues to develop pipeline in the target countries and invests in 8-12 projects.	1.1) Project identification and screening based on Arbaro's investment strategy climate eligibility criteria and ESMS 1.2) Due diligence on the potential investee projects 2.1) Approval of the proposed investments by the Arbaro Investment Committee 2.2) Execution of the investments	Funding is deployed. Project level operations commence.
3) Investments in project level operations	The investee companies commence the Arbaro funded operations.	Arbaro ensures that investment proceeds are used for the Arbaro approved operations such as: 4.1) Planting and forest management activities 4.2) Nursery operations 4.3) Harvesting, processing and wood sales as well as other portfolio company operations 4.4) Conservation activities	Carbon sequestration increases through the project activities. Employment is created

4) Arbaro Fund investment monitoring	Arbaro monitors the portfolio companies' operations to insure adherence to Arbaro' s investment strategy and ESMS.	Fund level: 3.1) Monitoring of portfolio company operations through reporting from the companies as well as active dialogue, including on-site visits, with the portfolio companies 3.2) Performance reviews to assess performance against targets	Annual reports to Arbaro fund investors on Arbaro fund's activities.
5) Reporting throughout the Arbaro Fund lifetime	Arbaro reviews the portfolio companies' performance against the set targets	Reporting: 5.1) Quarterly reports on the performance of the portfolio companies 5.2) Annual E&S and impact report	Annual reports to Arbaro Fund investors on Arbaro Fund's activities Annual Sustainability report
6) Exit from investments	Arbaro negotiates and executes the exit documentation. The exit proceeds are received.	6.1) Assessment of potential exit channels 6.2) Decision on final exit 6.3) Negotiation and execution of all exit documentation 6.4) Receipt and subsequent distribution of the exit proceeds	Final fund lifecycle target achievement on carbon sequestration, job creation, livelihood support and financial returns are realized and can be reported.

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Arbaro's reporting obligations are stipulated in the limited partnership agreement which is entered into between the Arbaro Fund and its investors. There may also be further reporting requirements under the accreditation master agreement which will be incorporated into the Arbaro Fund legal documentation, namely the side letter of the AE / GCF.

In addition, Arbaro provides comprehensive impact reporting.

The key reports that Arbaro prepares for its investors will include:

Quarterly reports (unaudited) which will be delivered no later than 60 calendar days after the end of each quarter and include:

- a summary of the key terms of the Arbaro Fund;
- a balance sheet of the Arbaro Fund, as of the end of such quarter;
- a year-to-date income statement of the Arbaro Fund;
- the amount of the manager remuneration (whether accrued and owing but not paid, or actually paid) paid by the Arbaro Fund to its manager (Arbaro Advisors) during the applicable quarter and fiscal year-to-date;
- a summary of the key developments in the preceding quarter (e.g. Drawdowns, new fund investments, investment pipeline, new investors, etc.);
- an overview of the Arbaro Fund investments, (i.e. commitments, stakes in the projects, drawdowns, IRR, valuation, compliance with the investment strategy) as well as a portfolio breakdown by country, counterparty and investment type; and
- a statement of changes in the relevant investor's capital account for the preceding quarter

Annual report (audited) which will be delivered four months after the end of each fiscal year and include a report audited by the Auditor setting forth:

- a balance sheet of Arbaro Fund as of the end of such fiscal year;
- a year-to-date income statement of Arbaro Fund for such fiscal year;
- a year-to-date statement of changes in the relevant investor's capital account for such fiscal year.

E&S and Impact reporting

The fund management team will report regularly and in a transparent manner about activities, results, and challenges of its business operations in line with its vision and mission. ESG issues shall be reported in depth to investors at least once a year, mainly through the publications of the Arbaro Sustainability Report. Furthermore, ESG key performance indicators will also be reported regularly. In years 5, 10 and 15 the report will include an assessment on whether similar initiatives have been developed in the regions of intervention, which would reflect the replicability of Arbaro activities.

Arbaro's Key Performance Indicators (KPIs) will be monitored and reported as follows:

- Financial: Arbaro follows the accounting standards of Lux GAAP. Net Asset Value calculation is performed quarterly and reported to the investors as part of the quarterly reporting
- CO_2 sequestration: Arbaro follows the overall guidance of the Intergovernmental Panel on Climate Change (IPCC) as well as methodologies from the Verified Carbon Standard (VCS). The carbon sequestered by Arbaro's projects corresponds to the net CO_2 sequestration as follows:

  The net CO_2 sequestration is estimated by applying the following equation:
  Net CO_2 sequestration (tCO_2) =
  CO_2 sequestration in the project – Baseline – Leakage – Other project emissions

- Job creation: Employment generation is measured as full-time equivalent. More detailed breakdowns of the employment generation will be available as well.

In addition, the annual sustainability report will provide details on the impact, final methodologies used will partly depend on each case and impacts thereof.
The Arbaro Fund performance will be compared to the Arbaro fund targets in mid-term and final assessment report based on Arbaro's KPIs.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Investments made into the Arbaro Fund are subject to market fluctuations and to the risks inherent in all investments. Accordingly, no assurance can be given that the investment objective of Arbaro will be achieved. The risk factors related to Arbaro can be categorized as fund risks and as risks related to investing in emerging markets and in the forestry sector. No exhaustive list of these risks can be given, and the risks may result in substantial net asset value volatility and depreciation or not achieving the impact targets of the fund.

General fund risks include risks related to individual investors, the fund structure and how it operates, and external environment related to both, e.g.:

- Currency risk (be it between individual investor's home currency and the fund currency; or the target project currencies)
- Changes is applicable law
- Reliance on the investment committee and investment managers
- Risk of non-payment by investors
- Use of leverage
- Dependency of key personnel

The risk mitigation strategy of the Arbaro Fund is pursued at both project and Fund level:

- Cautious and highly professional project planning and implementation, in strict compliance with FSC and IFC standards, allows for risk mitigation at project level
- Balanced project portfolio in terms of investment types (Greenfield, Brownfield, downstream), regions and products, allows for risk diversification at Fund level

Specific risk factors are continuously assessed throughout the entire investment process and are strictly monitored. Appropriate mitigation strategies are analyzed and implemented proactively from the earliest stage possible.

In the below section the key risk factors related to the investee projects, the investment operations in emerging markets and forestry sector and the mitigation measures to those are discussed in more detail.

Selected Risk Factor 1	**Biotic Risks**	
Category	Probability	Impact
Technical and operational	High	Low
Description		
In plantation investments, inadequate planning and implementation of silvicultural measures, as well as the occurrence of pests and diseases, can result in poor growth and quality of wood products. Since wood products are the major source of revenues for these investments, the underperformance of plantations will have a direct negative impact on the financial performance of the investment		
Mitigation Measure(s)		
Plantation investments will be conducted according to best silvicultural practices. This starts in the planning phase, with proper selection of species for the given site conditions, definition of silvicultural operations for selected species and production targets, and acquisition of high-quality planting materials; and continues throughout the production cycle through the timely implementation of silvicultural operations, conducted by skilled staff under close supervision. Proper planning and implementation of operations will minimize deviations from the plan of plantation performance and will reduce the vulnerability of plantations against any pests and diseases. In addition, pest and disease management programs will be put in place, including preventive measures such as species diversification. Finally, plantation investments will be strictly monitored to ensure early detection of pests, diseases and performance deviations, making it possible to take prompt measures whenever necessary.		

Selected Risk Factor 2	**Abiotic Risks**	
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
In addition to biotic factors described above, abiotic calamities, such as floods, storms, and fire events, can have a negative impact on the performance of plantations, and thereby on the financial performance of the investments. Such events can weaken plantations, making them more vulnerable against future events, pests and diseases, and even result in the loss of part of the planted area.		
Mitigation Measure(s)		
Plantation investments will be managed in a way that they improve landscape resilience. Management practices such as species diversification and high-quality silvicultural management will reduce the vulnerability of plantations against extreme climate events that might occur in the region, thus reducing any negative impacts on their performance. Fire events need special consideration in plantation investments. Forest fires can cause the loss of significant planted areas and in some cases, they are the result of social conflicts and agricultural practices in surrounding areas, since fire a common practice used to prepare agricultural and pasture land. Proper fire management strategies will be put in place, focusing on preventive measures and contingency plans. Fire prevention measures are integrated in technical operations, by ensuring an adequate level of diversification and avoiding large areas of homogenous vegetation, as well as conducting maintenance operations in a timely manner. Other measures include the construction and maintenance of the firebreak network, providing regular training and equipment for firefighting to employees, implementation of procedures for early detection and emergency plans, monitoring of fire events and record of statistics. Sufficient resources for fire management must be planned in annual budgets. The risk of fire events will be managed as well through proactive social strategies, targeting cooperation and aligning interests with local communities, through employment generation, enhancement of livelihoods and promotion of out-grower schemes, as well as fire awareness campaigns.		
Selected Risk Factor 3	**Global market risks**	
Category	Probability	Impact
Technical and operational	Medium	Low
Description		
Unfavorable changes in forest product markets, namely a decrease in demand and product prices, can compromise the financial performance of plantation investments. Global trends indicate that the probability of such events is low. Population growth and economic development, together with increasing wood supply gaps, suggest that favorable market conditions will persist.		
Mitigation Measure(s)		
Arbaro will mainly target countries that face a significant gap in sustainably produced industrial roundwood. Forecasts on population growth and economic development in target regions anticipate increasing product demand. Along with this, deforestation pressures and insufficient reforestation efforts suggest that this situation will become more critical in the future, resulting in severe problems for domestic forest industries. Furthermore, forest investments will be developed in regions with good growing conditions, ensuring the cost competitiveness of the wood produced. Arbaro will seek an adequate level of diversification in its portfolio in terms of species, forest products and domestic / export markets, as well as a balance between emerging / semi-mature forest markets.		
Selected Risk Factor 4	**Market risks at project level**	
Category	Probability	Impact
Technical and operational	High	Medium

Description
Beyond global forest product market trends, the decrease in demand or wood prices in the target market of a specific plantation investment will directly damage its financial performance. Also, operational issues with service providers, on which the timely and proper implementation of operations depends, and with clients, that ensure successful marketing of products, will have an impact on the long-term viability of the investment.
Mitigation Measure(s)
Plantation investments will be planned and implemented following a strict site-species-market approach. Market intelligence will be an essential part of the planning process, through species selection according to target products and markets, assessment of market access and deep understanding of forest industries and competitive landscape. Market development will be strictly monitored. Alternative market options will be assessed in all investments, ensuring a certain level of flexibility in case of unexpected market changes. Arbaro pursues long-term sustainable business, for which a high level of satisfaction among employees and customers is crucial. For this, investments will be managed by highly professional teams, will generate formal employment and will create high-standard working conditions.

Selected Risk Factor 5	**Social and Environmental Risks**	
Category	Probability	Impact
Governance	Medium	Medium
Description		
<u>Pests and diseases</u> Many pest and disease problems in plantations arise from the nature of forest operations, and not directly from growing trees in plantations. Planting of one species, whether indigenous or exotic, inevitably results in some areas where trees are ill suited to the site and suffer stress. This sometimes occurs in large-scale planting programmes where insufficient attention is paid to sites or where exotics are used extensively before sufficient experience has been gained. Also poor establishment and maintenance of plantations will increase their susceptibility to pests and diseases. <u>Fire and Storms</u> Plantation uniformity possibly increases risk from hurricane and storm damage if only because trees may be planted in locations that increases their susceptibility. This is particularly true at young ages and when plantations are not properly managed. These risks are best mitigated by careful planning and site species matching and by applying best silviculture practises. Finally, depending on location these risks can sometimes be insured. Most forest fires in plantations are caused by arson; only a few by lightning or encroachment of fires from neighbouring land. While there are a few examples of frequent fires preventing plantation development, it is more often due to poor community relationships than any inherent shortcoming with forest plantations. Fire risk is considered as one of the most important risks due to its potentially devastating impact. <u>Poor stakeholder relations</u> Poor stakeholder relations often develop from misinterpreting who the stakeholder groups are, poor communication and practises that are not transparent. At its worst, poor relations may lead to significantly increased fire risk in plantations due to arson or the project losing its social license to operate.		
Mitigation Measure(s)		
<u>General</u> Arbaro will select and manage investments according to its ESG policy and guidelines. IFC performance standards will be strictly applied in all projects to assess and effectively address potential social and environmental risks. All investments will be FSC certified. Arbaro will apply a rigorous internal red flag system and will involve communities to		

support development and reduce risks. Environmental and social risk mitigation strategies will be applied in all projects and they include:

- No establishment of plantations in natural forests
- Strict site-species matching ensures planting only in suited areas (suited biophysics, particularly annual precipitation > 1,000 – 1,200 mm; no competition with other land uses)
- Professional planning and site selection, including demarcation and management of conservation areas
- Professional application of best practices in forest operations, minimizing environmental impacts
- Risk management strategy related to genetic variability and fire prevention
- Transparent communication and participative relations with local communities and stakeholders
- Monitoring and evaluation of plantation performance, social and environmental impacts
- FSC certification, involving external audits and continuous improvement of applied practices

Pests and diseases

These risks are mitigated by careful planning of silvicultural management, site species matching and by applying best silviculture practises (e.g. diversifications of species, high-quality planting materials, timely execution of operations by well-trained staff).

Fire and storm damage

In all Arbaro's projects fire prevention measures, such as firebreaks in the plantations, monitoring and early detection, training as well as pre-agreed action plans in case of such occurrence are applied. Arbaro projects will also maintain good relations with the surrounding communities, with transparent practises and open communication. In addition to providing significant employment opportunities in their area of operation, Arbaro's projects will also actively educate the local communities of the benefits of forestry and promote out-grower and small-holder schemes where local communities can participate in the sector

Poor stakeholder relations

Arbaro projects will apply participatory approach to make sure that the local communities are well informed and involved in the project development. In identifying relevant stakeholders and assessing various approaches to be taken, Arbaro will follow the guidelines of the IFC Performance Standards, IFC's Stakeholder Consultations -Good Practises Handbook as well as the recommendations of the FSC.

Selected Risk Factor 6	**Political and FX risks**	
Category	Probability	Impact
Other	Medium	Medium
Description		

Expropriation

Expropriation or other such political risk event is considered extremely unlikely.

Corruption

Arbaro will implement an absolute zero tolerance to corruption. This may mean in the short term that e.g. licencing or approvals are delayed or other such administrative stumbling blocks will appear, however previous experience has evidenced that over medium term, zero tolerance policy will gradually remove these practises from the vicinity of zero-tolerance projects / companies.

Transfer restrictions / currency inconvertibility

Due to long term nature of forestry investments these risks could materialise

Legal risks

Inadequacy of local regulatory environmental.

Mitigation Measure(s)

Political risks can to an extent be mitigated by making sure that the projects are implemented in a transparent manner and making sure that the local stakeholders (from grassroots level to local and national government level) are appropriately informed and heard of throughout the implementation process. In addition, the mitigation measures include local participation schemes, integration of neighbouring communities as well as good cooperation with the local administration.

Risk related to transfer restrictions can be mitigated by timber sales in hard currency, possibility to speed up/reduce harvesting during challenging times as well as structuring investment via politically stable jurisdictions.

Arbaro's primary target countries are known to the management team from previous experience, and hence it is also known that the basic regulatory environment exists. Regardless of level of regulation, Arbaro's projects will always be implemented following the highest industry standards, including those related to environmental and social aspects of the projects. However, it is possible that either with time there will be adverse developments in regulation or that other players, e.g. competitors take advantage of such lacking regulation and through this enjoy competitive advantages. This could, over time, put Arbaro's project in adverse situation. To mitigate these Arbaro will work closely and transparently with local stakeholders including the local administration to communicate the benefits of the high standards followed. Arbaro's projects will also participate in e.g. local lobby groups such as industry associations to influence regulatory development as well as educate the decision makers on the industry.

G. GCF POLICIES AND STANDARDS

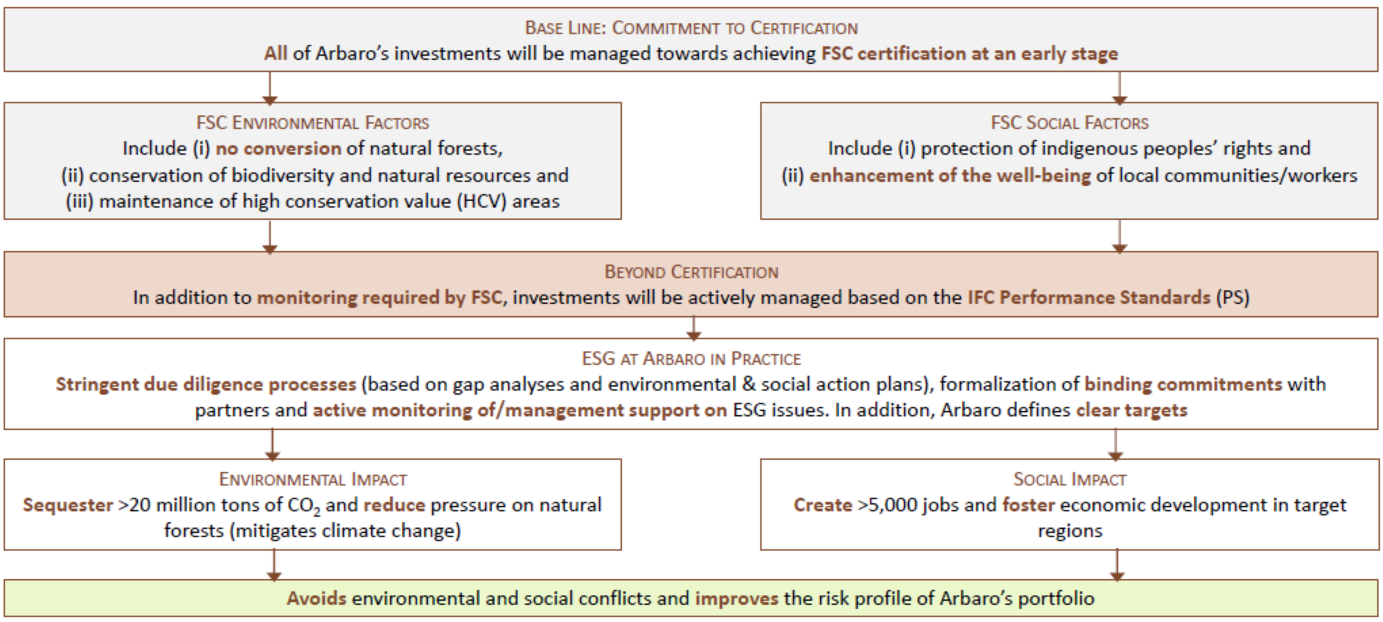
G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Arbaro's approach to E&S matters is governed by its ESMS. This is based on the International Finance Corporation (IFC) Policy and Performance Standards on Social and Environmental Sustainability (IFC PS), the related Environmental, Health and Safety Guidelines (EHS Guidelines) and the European Investment Bank's Environmental and Social Handbook. In addition, the Arbaro's E&S guidelines are compatible with the United Nations Principles for Responsible Investment (PRI).

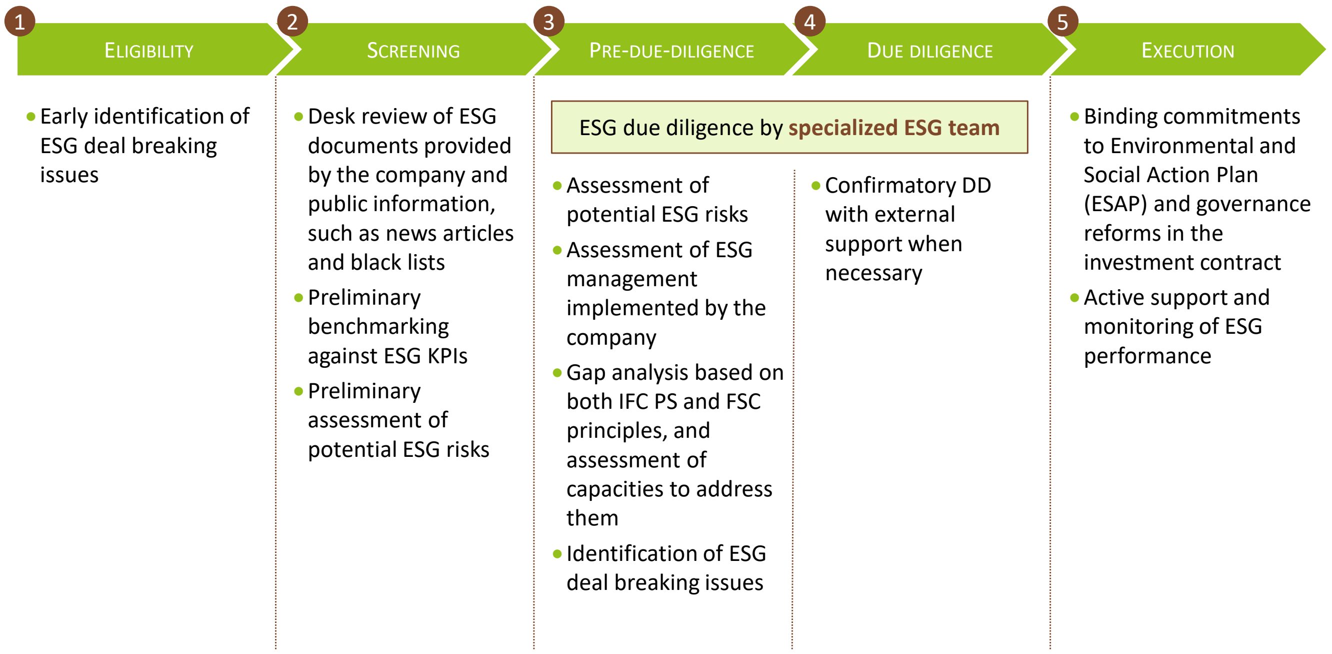
General principles of these E&S guidelines are:

- Finance only business operations that can comply with all applicable local and national laws, as well as international conventions and agreements ratified by the host country
- Operate according to the mitigation hierarchy, seeking to avoid, and where this is not possible, minimize adverse impacts and enhance positive effects on the environment, workers and stakeholders
- Actively seek for senior level endorsement of, and adherence to E&S guidelines of financed businesses, establishing formal commitments through appropriate E&S policies and practices, targets, budgets and timetables for implementation
- Integrate its E&S principles early in the planning process of each investment project in order to establish management systems subject to continuous improvement which effectively address E&S risks and realize E&S opportunities
- Set in place appropriate indicators and realize the follow-up of E&S measures throughout the investment process
- Promote and work towards the implementation of relevant international best practice standards, particularly FSC certification

Overview of Arbaro's ESG framework:



The E&S principles are a fundamental part of the Arbaro's business operations and compliance will be achieved through the means of an appropriate E&S management system. An effective E&S management system requires E&S principles to be respected from the initial phase of the investment process throughout the entire project cycle as described in the below graph:



ESIAs, as well as assessments related to resettlement or indigenous peoples will be conducted when and if applicable on a project level. As the projects are only chose during the implementation period (the investment period of the fund) such documents and assessments are not yet available. However, attached are the relevant policies and fund level documents [see Annex 6].

The programme will develop 8-12 projects that may fall under different risk categories:

- Category A: Potential significant adverse environmental or social risks and/or impacts that are diverse, irreversible, or unprecedented
- Category B: Potential limited adverse environmental or social risks and/or impacts that are few in number, generally site-specific, largely reversible, and readily addressed through mitigation measures
- Category C: Minimal or no adverse environmental or social risks and/or impacts

Given that projects are not defined at this stage, the overall programme is rated as Category A which is the highest possible risk category of projects.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

Gender equality is especially important factor to consider in forestry, as it has traditionally been male-dominated industry. Arbaro is committed to promoting gender equality and the basis for Arbaro's Gender Policy is formed by the IFC's Sustainability Framework and the Performance Standards as well as the FSC standards, both of which have specific sections on gender equality. The Fund seeks to promote gender equality throughout the whole investment process in four thematic areas:

- True participation: Arbaro recognizes that minimum number of women represented is not always enough, but their actual participation and ability to influence is more relevant.
- Equal benefits: Arbaro will use its best endeavors to ensure that the social and economic benefits that its stakeholder groups derive from its operations will be distributed in an equal basis considering the needs of the various groups and noting that such needs may vary between the groups.
- Capacity development: Arbaro is committed to contribute to capacity development as part of its activities and will seek to understand the specific needs of the target groups of these activities to understand their usefulness, and where applicable organize trainings to empower women and strengthen their capacities.
- Gender-disaggregated data: The Fund will seek to develop and apply transparent assessment and reporting methods that will include gender-disaggregated data. The objective of this data is to be able to provide meaningful information and regular feedback on the impact and effectiveness of the fund´s gender-responsive approach.

Project level gender assessment can only be provided at a later stage and on a project to project basis as the target projects will only be chosen during the implementation period (the fund investment period).
When assessing project opportunities Arbaro will seek to understand the prevailing socio-economic and cultural context as well as gender norms and perceptions related to the project area and all of its stakeholders. Where applicable, a gender-responsive social analysis as part of the overall social and environmental impact assessment will be undertaken to understand the underlying causes and drivers of gender inequalities.
Where laws and regulations disadvantage one gender over the other (often women, but men can also be affected), the Arbaro will seek to ensure that disadvantaged populations are protected and not further disadvantaged.
Where applicable, Arbaro will seek to address systemic and structural practices that create barriers to the realization of women's rights and gender equality.
Arbaro's investee companies will be encouraged to provide and operate in an environment that aspires to creating equal opportunities for both women and men, which in some instances may result in going "beyond the law". Investee companies will be required to have a gender policy, and a timetable and procedures for monitoring its implementation. Gender equality is included as a company performance indicator.
For details please refer to Annex 8.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

The Fund Manager, FiM Asset Management S.à r.l. is registered with the Luxembourg financial sector regulator CSSF as an Alternative Investment Fund Manager (AIFM).

EY Luxembourg has been chosen as the auditor of the Fund.

Alter Domus is the selected as administrative and depositary agent for the Fund. Alter Domus is a leading provider of Fund and Corporate services. Established in 2003 its 1,100 employees serve most of the largest debt, real estate and private equity fund in the world. Alter Domus is certified according to SSAE AT Section 801 / ISAE 3402 SOC 1 Type II.

The annual accounts of the Fund will be audited as of the end of each fiscal year by the Fund's independent auditor. All annual accounts provided to the Limited Partners will be prepared in accordance with Lux GAAP.

The valuation approach of the Fund follows the general principles of the International Private Equity and Venture Capital Valuation Guidelines dated December 2015 (as amended from time to time) published by Invest Europe

Other procurement

The major part of the investment process will be performed by the Arbaro Advisors team itself. Contrary to many other timber funds, the Arbaro team will also perform the technical due diligence internally. Arbaro will be supported by transaction support services provided by Finance in Motion and UNIQUE at no additional costs to the Fund, which is why the use of third-party external consultants and professional services firms will be limited.

Any external services, such as the legal, tax, audit or technical service, will be procured in a professional and commercial manner based on well-defined terms of references, competitive rates, monitoring and control of service delivery, and evaluation for future assignments. Where possible, the Fund aims to efficiently work together with a recurring panel of external service providers.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒ Annex 1 NDA no-objection letter(s) **(template provided)**

☐ Annex 2 Feasibility study - and a market study, if applicable

☐ Annex 3 Economic and/or financial analyses in spreadsheet format

☒ Annex 4 Detailed budget plan **(template provided)**

☒ Annex 5 Implementation timetable including key project/programme milestones **(template provided)**

☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)**

☐ Environmental and Social Impact Assessment (ESIA) or

☐ Environmental and Social Management Plan (ESMP) or

☐ Environmental and Social Management System (ESMS)

☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)

☐ Annex 7 Summary of consultations and stakeholder engagement plan

☒ Annex 8 Gender assessment and project/programme-level action plan **(template provided)**

☐ Annex 9 Legal due diligence (regulation, taxation and insurance)

☐ Annex 10 Procurement plan **(template provided)**

☐ Annex 11 Monitoring and evaluation plan **(template provided)**

☐ Annex 12 AE fee request **(template provided)**

☐ Annex 13 Co-financing commitment letter, if applicable **(template provided)**

☐ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☐ Annex 15 Evidence of internal approval **(template provided)**

☐ Annex 16 Map(s) indicating the location of proposed interventions

☒ Annex 17 Multi-country project/programme information **(template provided)**

☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project

☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity

☐ Annex 20 First level AML/CFT (KYC) assessment

☐ Annex 21 Operations manual (Operations and maintenance)

☒ Annex 22 Carbon Methodology

☒ Annex 23 Sustainable Forestry

☒ Annex 24 Country Context

☒	Annex 25	Climate Eligibility
☐	Annex 26	LPA